Geethanjali College of Engineering and Technology

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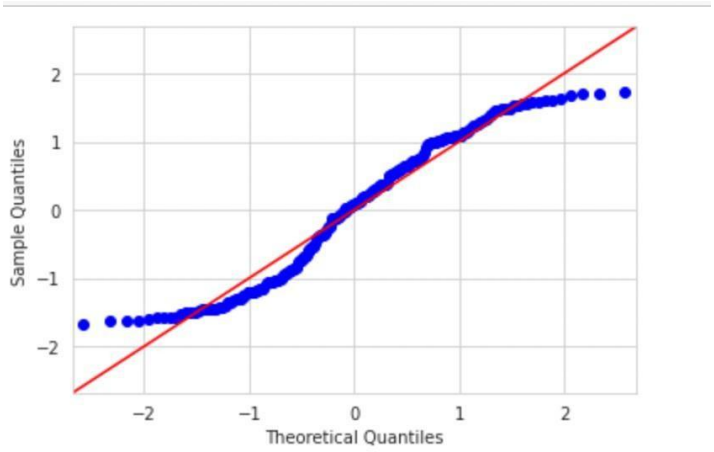
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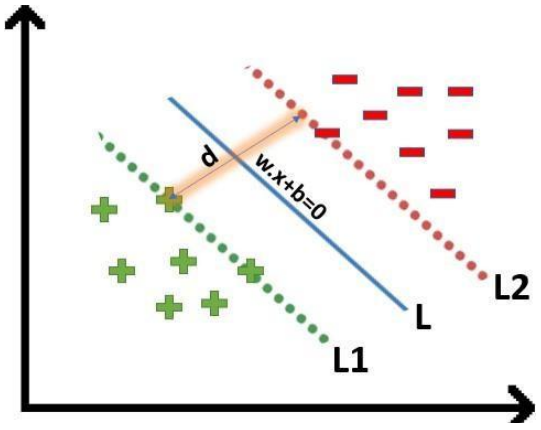
STATISTICS FOR MACHINE LEARNING

LABORATORY WORKBOOK

III Year B. Tech-I Semester (AY 2025-26)







Name: …….....Roll No: .....……

Branch: …….....Section: ……....

Geethanjali College of Engineering and Technology

AUTONOMOUS

(Accredited by NBA, NAAC "A+" Grade)

Cheeryal (V), Keesara (M), Medchal District – 501 301 (T S)



STATISTICS FOR MACHINE LEARNING LAB

Subject Code – (20MA31L01)

LABAORATORY MANUAL

III Year B.Tech I Semester

Common to

MECHANICAL ENGINEERING

2025-2026

DEPARTMENT OF MECHANICAL ENGINEERING

Geethanjali College of Engineering and Technology

AUTONOMOUS

(Accredited by NBA, NAAC "A+" Grade)

Cheeryal (V), Keesara (M), Medchal District – 501 301 (T S)



CERTIFICATE

This is to certify that Mr. / Miss _____

has satisfactorily completed _____*number of programmes in the* ***Statistics for Machine Learning lab****.*

Roll No: _____ Branch: _____

Year : _____ AcademicYear: _____

Head of the Department Faculty In-charge

Internal Examiner External Examiner

Geethanjali College of Engineering and Technology
(Autonomous)
DEPARTMENT OF MECHANICAL ENGINEERING

(Name of the Subject/Lab Course): STATISTICS FOR MACHINE LEARNING
(Subject Code): 20MA31L01 **Programme : UG**

Branch : ME **Version No : 1**
Year : III **Updated on : 06/0/2025**

Semester : I **No. of pages : 88**

Classification status (Unrestricted / Restricted)
Distribution List: Department, Lab, Lab In Charge

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2) Sign :
3) Date :

Geethanjali College of Engineering and Technology
AUTONOMOUS

Subject code: 20MA31L01

L	T	P/D	C
-	-	**2/-**	1

Pre-requisite(s): 20MA31001- Statistics for Machine Learning

Course Outcomes:
At the end of the course, student would be able to use R software to:

CO1: Construct frequency distribution for the given data.

CO2: Obtain specified set of samples from large data through random variables using Acceptance-Rejection (AR) algorithm.

CO3: Estimate and test the significance of the specified population parameter(s) for the given data.

CO4: Reduce the dimensionality of large data sets using principal component analysis (PCA),Logistic Regression(LR),K-means clustering and Support Vector Machine(SVM) for selecting variables

List of Experiments:

1. Introduction to R Programming
2. Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)
3. Acceptance/Rejection Sampling in R
4. Maximum likelihood and method of moments estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R
5. Testing of hypothesis based on t-test, chi-square test and F-test, Confidence interval estimation in R
6. Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R
7. Logistic Regression in R
8. Classification with SVM in R
9. Dimensionality reduction with PCA in R and K-means Clustering in R
10. Kernel PCA and Kernel SVM in R

List of Additional programs

1. Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R Programming
2. Simple Linear Regression :Testing overall hypothesis and testing significance of variables
3. Multiple linear regression: Testing overall hypothesis and testing significance of individual variables, model selections and prediction in R programming.

Vision of the Institute:

Geethanjali visualizes dissemination of knowledge and skills to students, who would eventually contribute to the well-being of the people of the nation and global community.

Mission of the Institute:

- To impart adequate fundamental knowledge in all basic sciences and engineering, technical and Interpersonal skills to students.
- To bring out creativity in students that would promote innovation, research and entrepreneurship.
- To Preserve and promote cultural heritage, humanistic and spiritual values promoting peace and harmony in society.

Vision of the Department:

Department of ME

The Mechanical Engineering department strives to be recognized globally for outstanding education and research, imparting quality education, churning well-qualified engineers, who are creative, innovative, and entrepreneurial, solving problems for societal development.

Mission of the Department:

Department of ME

- Imparting quality education to students to enhance their skills and make them globally competitive.
- Prepare graduates to engage in life-long learning, possess intellectual capabilities, serving society with a strong commitment to their profession, meeting technical challenges and exhibiting ethical responsibility for societal development. To be a center of excellence in instruction, innovation in research and scholarship, and service to the stake holders, the profession, and the public.

- Conduct quality research, create opportunities for students and faculty to showcase their talent, disseminate knowledge, and promote community development, leading to peace and harmony in society.

Program Educational Objectives (PEO's):

ME

PEO 1: To prepare students with strong fundamental knowledge in basic sciences mathematics, engineering courses which would facilitate them find gainful employment with a sense of appreciation to pursue life-long learning for professional development.

PEO 2: To inculcate problem solving skills in students, imbibing creativity and innovation which would enable them to develop modern machinery involving cutting edge technologies of Multidisciplinary nature for societal development.

PEO 3: To develop critical thinking with an aptitude to conduct research and development, instill professional ethics, develop effective communication and interpersonal skills with positive attitude to contribute significantly towards their chosen profession, thereby supporting community development.

Program Outcomes (POs):

1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

2. **Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

3. Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

4. Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

5. Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

6. The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

8. Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make

effective presentations considering cultural, language, and learning differences

10. Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. Life-Long Learning: Recognize the need for, and have the preparation and ability for **i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)**

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost,re- use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in theengineering discipline.

WK7: Knowledge of the role of engineering insociety and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer topublic safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding andrespect, and of inclusive attitudes.

Program Specific Outcomes (PSOs):

ME

PSO 1: Apply Continuity, Energy and Momentum equations to mechanical systems, design and perform experiments in all fields of mechanical engineering

PSO 2: Able to analyze, design and develop/model mechanical and its allied systems using software tools such as AUTOCAD, ANSYS, Creo etc.

PSO 3: Able to design layouts for process and manufacturing industry taking into consideration optimization of resources for effective operation and maintenance.

Mapping of Lab Course outcomes with Program outcomes:

ME

CO-PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PS O1	PS O2	PS O3
CO1	3	2	-	-	-	-	-	-	-	2	-	2	-	-
CO2	3	2	-	-	-	-	-	-	-	2	-	2	-	-
CO3	3	2	-	3	-	-	-	-	-	3	-	3	-	-
CO4	3	2	3	3	-	-	-	-	-	3	-	3	-	-

-	No Correlation	1	Slight (low)	2	Moderate (medium)	3	Substantial (High)

Prerequisites:

Statistics for Machine Learning (20MA31001)

INSTRUCTIONS TO THE STUDENTS:

1. Students are required to attend all labs.
2. Students should be dressed in formals when attending the laboratory sessions.
3. Students will work individually in computer laboratories.
4. While coming to the lab bring the observation book and Work book etc.
5. Before coming to the lab, prepare the pre-lab questions. Read through the lab experiment to familiarize you.
6. Utilize 3 hours' time properly to perform the experiment and noting down the outputs.
7. If the experiment is not completed in the prescribed time, the pending work has to be done in the leisure hour or extended hours.
8. You will be expected to submit the completed work book according to the deadlines set up by your instructor.

INSTRUCTIONS TO LABORATORY TEACHERS:

1. Observation book and lab records submitted for the lab work are to be checked and signed before the next lab session.
2. Students should be instructed to switch ON the power supply after the connections are checked by the lab assistant / teacher.

3. The promptness of submission should be strictly insisted by awarding the marks accordingly.
4. Ask Viva Voce questions at the end of the experiment.
5. Do not allow students who come late to the lab class.
6. Encourage the students to do the experiments innovatively.
7. Fill Continuous Evaluation Sheet, on regular basis.
8. Ensure that the students are dressed in formals

Evaluation pattern:

Total Lab Marks –100 (Internal - 40 Marks & External – 60 Marks)

Internal Assessment:

1. Continuous valuation is done for 15 Marks based on:
   a. Submission of workbook in time (5 Marks)
   b. Performance of Program (5 Marks)
   c. Viva-Voce (5 Marks)
2. During the semester, two Lab Internal examinations are conducted for 15 Marks and average will be considered.
3. Project presentation Seminar conference Presentation carries 10 marks

External Assessment:

At the end of semester, external lab exam is conducted for 60 Marks with one external examiner and one internal examiner.as per regulation.

- Write up and program code (25 Marks)
- Execution of Program (25 Marks)
- Viva-Voce (10 Marks)

PERFORMANCE INDICATOR

S No.	Name of the Program	Page No.	Date of Programme	Date of Submission	Marks awarded	Faculty Sign.	Remarks
1	Introduction to R Programming						
2	Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)						
3	Acceptance/Rejection Sampling in R						
4	Maximum likelihood and method of moments estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R						
5	Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R						
6	Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R						
7	Logistic Regression in R						
8	Classification with SVM in R						
9	Dimensionality reduction with PCA in R and K-means Clustering in R						
10	Kernel SVM in R						

	ADDITIONAL PROGRAMMES						
S No.	**Name of the Program**	**Page No.**	**Date of Programme**	**Date of Submission**	**Marks awarded**	**Faculty Sign.**	**Remarks**
1	Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R programming						
2	Simple Linear Regression in R programming.						
3	Multiple linear regression: Testing over all hypothesis and testing significance of individual variables, model selections and prediction in R programming						

S No.	INDEX	Page No.
1	Introduction to R Programming	
2	Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)	
3	Acceptance/Rejection Sampling in R	
4	Maximum likelihood and method of moments estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R	
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	Additional Programmes	
1	Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R programming	
2	Linear Regression :Testing overall hypothesis and testing significance of Variables	
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1. Introduction to R Programming

R is a programming language and free software environment for statistical computing and graphics that is supported by the R Foundation for Statistical Computing. The R language is widely used among statisticians and data miners for developing statistical software and data analysis.

R is a programming language and software environment for statistical analysis, graphics representation and reporting.

The following are the important features of R

- R is a well-developed, simple and effective programming language which includes conditionals, loops; user defined recursive functions and input and output facilities.
- R has an effective data handling and storage facility,
- R provides a suite of operators for calculations on arrays, lists, vectors and matrices.
- R provides a large, coherent and integrated collection of tools for data analysis.
- R provides graphical facilities for data analysis and display either directly at the computer or printing at the papers.

To Install R and R Packages:

1. Open an internet browser and go to www.r-project.org.
2. Click the "download R" link in the middle of the page under "Getting Started."
3. Select a CRAN location (a mirror site) and click the corresponding link.
4. Click on the "Download R for WINDOWS" link at the top of the page.
5. Click on the file containing the latest version of R under "Files."
6. Save the **.pkg** file, double-click it to open, and follow the installation instructions.
7. Now that R is installed, you need to download and install R Studio**.**

To Install R Studio:

1. Go to www.rstudio.comand click on the "Download R Studio" button.
2. Click on "Download R Studio Desktop."
3. Click on the version recommended for your system, or the latest Mac version, save the .dmg file on your computer, double-click it to open, and then drag and drop it to your applications folder.

Install R Packages:

1. Download and install a package (you only need to do this once).

2. To use the package, invoke the library (package) command to load it into the current session. (You need to do this once in each session, unless you customize your environment to automatically load it each time).

Installing and Loading Packages:

R to load the package's library in R session

> library ("MASS") #R now knows all the functions that are scanned in the MASS package

1.1. Basics and R as a Calculator:

i. 31 * 78 # Multiplication

[1] 2418

ii. 697/41 #division

[1] 17

iii. Assign the value of 39 to x

x<-39

> x

[1] 39

iv. Assign the value of 22 to y

> y<-22

> y

[1] 22

v. Make z the value of x - y

> z<- x - y

vi. Display the value of z in the console

> z

[1] 17

vii. Find 10 random numbers between 1 and100. (Hint: you can use sample() function)

> sample(1:100,10)

[1] 77 74 89 23 39 84 63 12 91 16

1 Example: Find the value of 21 is multiplied by 23, 32 is multiplied by 62; 1256 divided by 2; Assign the value of 26 to x variable, 33 to y variable and hence find the value of z such that z = x+y; x-y;

R Code with Output:

```
> 21*23
[1] 483
> 32*62
[1] 1984
> 1256/2
[1] 628
> x=26
> y=33
> z=x+y
> z
[1] 59
> x-y
[1] -7
```

2 Example: Find 5 random numbers between 1 to 100 using sample function

R Code with Output:

```
> sample (1:100,5)

[1] 23 94 30 62 45
```

1.2. Creating Vectors:

A vector is a basic data structure of R containing the same type of data.

R Code for vector calculations:

```
x = c(observations / data)   # Store the observations in x
x+y                          # Addition of two vectors
x-y                          Subtraction of two vectors
x*y                          Multiplication of two vectors
```

1. Example: The numbers below are the first ten days of rainfall amounts in 1996. Read them into a vector using the vector function named as "rainfall"

0.1, 0.6, 33.8, 1.9, 9.6, 4.3, 33.7, 0.3, 0.0, 0.1

R-Code with Output:

```
> rainfall = c (0.1,0.6,33.8,1.9,9.6,4.3,33.7,0.3,0.0,0.1)
> rainfall
 [1] 0.1 0.6 33.8 1.9 9.6 4.3 33.7 0.3 0.0 0.1
```

2. Example: Find the addition, subtraction and multiplication of two vectors

A= 20,22,24,25,21,26,25,26,21,25,26,24,25,25,26,25,24,25,26,25

B= 15,14,15,16,15,14,18,16,14,15,15,17,15,15,14,15,16,17,18,19

R-Code with Output:

A=c(20,22,24,25,21,26,25,26,21,25,26,24,25,25,26,25,24,25,26,25)

B=c(15,14,15,16,15,14,18,16,14,15,15,17,15,15,14,15,16,17,18,19)

```
A
B
```

```
A=c(20,22,24,25,21,26,25,26,21,25,26,24,25,25,26,25,24,25,26,25)
B=c(15,14,15,16,15,14,18,16,14,15,15,17,15,15,14,15,16,17,18,19)
```

A+B

```
> A+B
 [1] 35 36 39 41 36 40 43 42 35 40 41 41 40 40 40 40 40 42 44 44
```

A-B

```
> A-B
 [1]  5  8  9  9  6 12  7 10  7 10 11  7 10 10 12 10  8  8  8  6
```

A*B

```
> A*B
  300 308 360 400 315 364 450 416 294 375 390 408 375 375 364 375 384 425

  468 475
```

1.3. Creating Matrices:

To create matrices we will use the matrix() function. The matrix() function takes the following arguments:

- data an R object (this could be a vector).
- nrow the desired number of rows.
- ncol the desired number of columns.
- by row a logical statement to populate the matrix by either row or by column

R Code for Matrix Calculations:

```
A=matrix(c(data),nrow=no.of  rows,  ncol=no.ofcolumns,  byrow=TRUE)
                                                  # Store the data in matrix A
                                          A+B,  # Addition,
                                          A-B,     Subtraction
                                       A%*%B     Multiplication of two matrices,
                                          t(A)     Transpose of a matrix
                                 rankMatrix(A)  # Rank of a matrix
                                       eigen(A)  # Eigen values and eigen vectors
```

1. Example: If x is a matrix specified by the R command **x = matrix(nrow=3, ncol=3, data=c(1,8,2,1,5,4,3,9,6), byrow=TRUE),** then the output of the R command **x%*%t(x)** is?

R-Code with Output:

```
> x=matrix(nrow=3,ncol=3,data=c(1,8,2,1,5,4,3,9,6),   byrow  =TRUE)

> x%*%t(x)

     [,1] [,2] [,3]
[1,]   69   49   87
[2,]   49   42   72
[3,]   87   72  126
```

2. Example: If $A = \begin{bmatrix} 0 & 3 & 1 \\ 5 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ are two matrices then check whether the two matrices satisfy commutative property or not. And also find transpose of each matrix using R program.

a) R-Code with Output:

A=matrix(c(0,3,1,5,1,-1,1,2,1),nrow=3, ncol=3, byrow=TRUE)

A

B=matrix(c(0,0,1,0,1,0,1,0,1),nrow=3, ncol=3, byrow=TRUE)

B

A+B

```
A=matrix(c(0,3,1,5,1,-1,1,2,1),nrow=3, ncol=3, byrow=TRUE)
> A
     [,1] [,2] [,3]
[1,]    0    3    1
[2,]    5    1   -1
[3,]    1    2    1
> B=matrix(c(0,0,1,0,1,0,1,0,1),nrow=3, ncol=3, byrow=TRUE)
> B
     [,1] [,2] [,3]
[1,]    0    0    1
[2,]    0    1    0
[3,]    1    0    1
>A+B
     [,1] [,2] [,3]
[1,]    0    3    2
[2,]    5    2   -1
[3,]    2    2    2


B+A
     [,1] [,2] [,3]
[1,]    0    3    2
[2,]    5    2   -1
[3,]    2    2    2
```

A+B=B+A

Under addition A & B matrices satisfy Commutative property.

```
> A%*%B
     [,1] [,2] [,3]
[1,]    1    3    1
[2,]   -1    1    4
[3,]    1    2    2

B%*%A
     [,1] [,2] [,3]
[1,]    1    2    1
[2,]    5    1   -1
[3,]    1    5    2
```

Under Multiplication A & B matrices does not satisfy commutative property.

b) R-Code with Output:

t(A)

```
t(A)
     [,1] [,2] [,3]
[1,]    0    5    1
[2,]    3    1    2
[3,]    1   -1    1
t(B)

t(B)
     [,1] [,2] [,3]
[1,]    0    0    1
[2,]    0    1    0
[3,]    1    0    1
```

3. Example: Find the rank of a matrix A= $\begin{bmatrix} 3 & 2 & 4 & 0 & -4 \\ 2 & 1 & 2 & 1 & 5 \\ 0 & 0 & 3 & -2 & 1 \\ 4 & 5 & 1 & 1 & 2 \\ 2 & 1 & 5 & -1 & 0 \\ 1 & 2 & 2 & 2 & 1 \end{bmatrix}$

R-Code with Output:

A=matrix(c(3,2,4,0,-4,2,1,2,1,5,0,0,3,-2,1,4,5,1,1,2,2,1,5,-1,0,1,2,2,2,1),nrow=6,ncol=5, byrow=TRUE)

A

```
A
     [,1] [,2] [,3] [,4] [,5]
[1,]    3    2    4    0   -4
[2,]    2    1    2    1    5
[3,]    0    0    3   -2    1
[4,]    4    5    1    1    2
[5,]    2    1    5   -1    0
[6,]    1    2    2    2    1
```

```
print(rankMatrix(A))
```

```
[1] 5
```

4. Example: Find the eigen values and eigen vectors of a given matrix $A=\begin{bmatrix}1 & 8 & 9 & 16\\ 2 & 7 & 10 & 15\\ 3 & 6 & 11 & 14\\ 4 & 5 & 12 & 13\end{bmatrix}$

R-Code with Output:

```
A=matrix(c(1,8,9,16,2,7,10,15,3,6,11,14,4,5,12,13),nrow=4,ncol=4,byrow=TRUE)

A
```

```
A=matrix(c(1,8,9,16,2,7,10,15,3,6,11,14,4,5,12,13),nrow=4,ncol=4,byrow=TRUE)
> A


     [,1] [,2] [,3] [,4]
[1,]    1    8    9   16
[2,]    2    7   10   15
[3,]    3    6   11   14
[4,]    4    5   12   13


print(eigen(A))


eigen() decomposition
$values
[1]  3.400000e+01 -2.000000e+00 -2.608772e-15 -5.588953e-16

$vectors
     [,1]        [,2]       [,3]        [,4]
[1,] -0.5 -0.82323920  0.6740714  0.82533832
[2,] -0.5 -0.42274445  0.6563350  0.33233043
[3,] -0.5 -0.02224971 -0.2483414 -0.45488432
[4,] -0.5  0.37824504 -0.2306049  0.03812357
```

Viva Questions:

1. Write the syntax for installing package MASS?

Ans. install.packages("MASS")

2. Write the syntax to load MASS package in library?

Ans. Library(MASS)

3. Write the R code to find the value of Z = X+Y. where X=26, Y = -10.

Ans. Z = 16

1.4. Sequence & Drawing a Scatterplot

1. Example: Print Numbers from 0 to 5 by the difference 0.5

R-Code with Output:

seq(from = 0, to = 5, by = 0.5)

```
seq(from = 0, to = 5, by = 0.5)
 [1] 0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0
```

2. Example: carage :5,7,8,7,2,2,9,4,11,12,9,6

Carspeed :99,86,87,88,111,103,87,94,78,77,85,86

Show these two variables in scatter plot and identify is there any relationship between these two variables.

R-Code with Output:

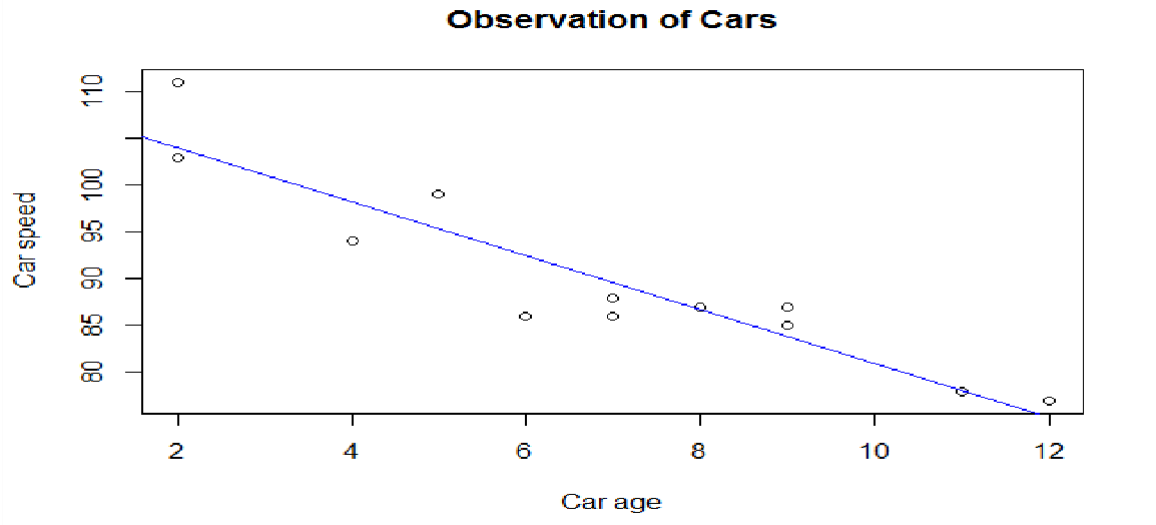
carage<- c(5,7,8,7,2,2,9,4,11,12,9,6)

carspeed<- c(99,86,87,88,111,103,87,94,78,77,85,86)

```
carage
 [1]  5  7  8  7  2  2  9  4 11 12  9  6
carspeed
 [1]  99  86  87  88 111 103  87  94  78  77  85  86
```

plot(carage, carspeed, main="Observation of Cars", xlab="Car age", ylab="Car speed")

abline(lm(carspeed ~ carage), col = "blue")



The **x-axis** shows how old the car is and **y-axis** shows the speed of the car when it passes.

It seems that the newer the car, the faster it drives.

2. Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)

Absolute Frequency, Relative Frequency:

The number of observations in a particular class / category is called the **Absolute Frequency** or **Frequency**.

The number of observations in a particular class / category is divided by total frequency is called **Relative Frequency**.

2.1. Absolute and Relative Frequency.

R Code:

table(data vector) #creates the *absolute frequency* of the **data vector** of the given data

Enter data as **x**

table(x) # displays Absolute Frequencies

table(x)/length(x) # displays Relative Frequencies

1. Example: Results of 10 persons declared in two categories as Pass (P) and Fail (F) is categorized as 1 and 2 respectively. Find Absolute and Relative Frequencies.

P, F, P, F, F, P, P, F, P, P

1, 2, 1, 2, 2, 1, 1, 2, 1, 1

R Code with Output:

```
> x=c(1,2,1,2,2,1,1,2,1,1)
> table(x)
x
1 2
6 4
> table(x)/length(x)
   x
   1   2
0.6 0.4
```

2. Example: The time (in minutes) taken to complete the puzzle was recorded as follows and stored in a data vector as puzzle time. Find Absolute and Relative frequencies.

8,13,21,9,15,9,21,13,8,8,13,15,8,21,9,15,8,13,21

R Code with Output:

```
> x=c(8,13,21,9,15,9,21,13,8,8,13,15,8,21,9,15,8,13,21)
> table(x)
x
 8  9 13 15 21
 5  3  4  3  4
```

```
> table(x)/length(x)
x
        8         9        13        15        21
0.2631579 0.1578947 0.2105263 0.1578947 0.2105263
```

2.2. Frequency Distribution: Arrangement of ungrouped data in the form of group is called Frequency Distribution of data.

Algorithm:

- First step is to find the "range" of the data values which can be partitioned into class interval.Use command range which returns a vector containing the minimum and maximum of all the given arguments.
- Create a sequence starting from lower limit to upper limit of given set of observations at an interval of width specified in the given problem.
- Now we need to convert Numeric to Factor using a command **cut**
- Now we can compute the absolute frequency of given data in each width interval with the **table()**
- Use the **cbind** function to print the frequency distribution in column format.
- To compute the relative frequency distribution of given data in each width interval with **table()** by **length**().

2.3. Cumulative Distribution Function (CDF) for data: It gives us an idea about the cumulative frequencies up to a certain point of the given data using **cumsum()**.

R Code:

>x=c(data vector)

>**range(x)** #returns a vector containing the minimum and maximum of all the given arguments.

>**breaks = seq(minimum value, maximum value, by=specified width)**

Sequence at interval of specified width.

>**C_I=cut(x, breaks, right = FALSE)** #divides the range of data vector into intervals and codes the values

in data vector according to which interval they fall

>**Freq=table(C_I)** # gives the absolute frequency of the variable of the data file which generates the

frequency distribution of the data on variable

>cbind(Freq) # print the frequency distribution in column format

>**Rel_Freq=table(C_I)/length(C_I)** #gives the relative frequency of the variable of the data file which generates the frequency distribution of the data on variable

>cbind(Rel_Freq) # print the relative frequency distribution in column format.

R Code for Cumulative Distribution function:

>**Cum_Freq=cumsum(Freq)** #returns a vector whose elements are the cumulative sums of the elements of the absolute frequency distribution.

>**cbind(Cum_Freq)** # print the cumulative frequency distribution in column format

Note: Suppose if we want to display the total Frequency distribution table consisting of Absolute Frequency, Relative Frequency and Cumulative Frequency in column Format then:

> cbind(Freq, Rel_Freq, Cum_Freq)

1. Example:

Following are the 50 test scores for male students in an examination.
79, 70, 85, 64, 41, 69, 66, 28, 89, 47, 88, 72, 21, 81, 75, 72, 55, 35, 56, 80, 68, 52, 49, 35, 20, 30, 23, 48, 90, 24, 58, 65, 29, 50, 67, 37, 42, 36, 86, 63, 78, 84, 60, 22, 57, 61, 32, 27, 73, 31.

a) Write R code to construct frequency distribution table of width 10.

b) And also find relative frequency distribution and cumulative frequency distribution for the male students.

c) Represent the above (a) and (b) in column format.

R code with Output:

x=c(79, 70, 85, 64, 41, 69, 66, 28, 89, 47, 88, 72, 21, 81, 75, 72, 55, 35, 56, 80, 68, 52, 49, 35, 20, 30, 23, 48, 90, 24, 58, 65, 29, 50, 67, 37, 42, 36, 86, 63, 78, 84, 60, 22, 57, 61, 32, 27, 73, 31)

```
x
 [1] 79 70 85 64 41 69 66 28 89 47 88 72 21 81 75 72 55 35 56 80 68 52 49 35
20 30 23 48
[29] 90 24 58 65 29 50 67 37 42 36 86 63 78 84 60 22 57 61 32 27 73 31
```

range(x)

```
range(x)
[1] 20 90
```

breaks = seq(20, 100, by=10)

breaks

```
breaks = seq(20, 100, by=10)
> breaks
[1]  20  30  40  50  60  70  80  90 100
```

C_I=cut(x, breaks, right = FALSE)

Freq=table(C_I)

Freq

C_I

```
C_I
  [20,30)   [30,40)   [40,50)   [50,60)   [60,70)   [70,80)   [80,90)  [90,100)
        8         7         5         6         9         7         7         1
```

Rel_Freq=table(C_I)/length(C_I)

Rel_Freq

```
Rel_Freq=table(C_I)/length(C_I)
> Rel_Freq
C_I
  [20,30)   [30,40)   [40,50)   [50,60)   [60,70)   [70,80)   [80,90)  [90,100)
     0.16      0.14      0.10      0.12      0.18      0.14      0.14      0.02
```

Cum_Freq=cumsum(Freq)

Cum_Freq

```
Cum_Freq=cumsum(Freq)
> Cum_Freq
  [20,30)   [30,40)   [40,50)   [50,60)   [60,70)   [70,80)   [80,90)  [90,100)
        8        15        20        26        35        42        49        50
```

cbind(Freq, Rel_Freq, Cum_Freq)

```
cbind(Freq, Rel_Freq, Cum_Freq)
          Freq Rel_Freq Cum_Freq
[20,30)      8     0.16        8
[30,40)      7     0.14       15
[40,50)      5     0.10       20
[50,60)      6     0.12       26
[60,70)      9     0.18       35
[70,80)      7     0.14       42
[80,90)      7     0.14       49
[90,100)     1     0.02       50
```

2. Example:

Following are the scores of 250 students in a test with minimum and maximum marks being 0 and 100 respectively. The scores are stored in a data vector named **scores**.

21.08, 13.76, 75.89, 68.41, 51.36, 21.58, 13.85, 82.75, 71.88, 57.93, 6.68, 24.99, 10.24, 24.60, 95.14, 99.86, 42.72, 3.66, 66.89, 96.60, 30.08, 49.50, 16.84, 89.14, 14.97, 66.86, 38.32, 71.90, 9.50, 66.92, 58.83, 38.06, 79.95, 95.30, 97.51, 49.67, 50.54, 93.64, 74.30, 76.82, 63.32, 36.54, 90.87, 72.37, 61.98, 89.44, 98.19, 27.07, 63.97, 55.35, 92.40, 32.97, 61.37, 34.40, 42.48, 77.67, 99.31, 54.64, 69.46, 49.88, 86.76, 31.95, 4.55, 36.77, 36.91, 56.45, 54.44, 76.53, 15.97, 18.22, 41.34, 50.17, 18.82, 96.60, 93.58, 8.85, 30.27, 44.18, 26.85, 54.82, 64.24, 46.20, 3.00, 96.14, 10.01, 75.64, 93.18, 93.32, 15.88, 84.66, 37.91, 97.11, 66.32, 87.91, 85.58, 64.32, 16.11, 37.22, 42.69, 89.64, 89.96, 48.29, 27.71, 89.71, 23.08, 53.02, 4.22, 18.84, 17.41, 73.05, 33.26, 71.83, 8.28, 69.65, 51.27, 75.61, 63.67, 8.24, 1.15, 83.05, 23.08, 26.37, 19.74, 34.79, 53.58, 66.88, 99.58, 63.30, 60.82, 43.71, 39.76, 54.97, 13.55, 81.42, 26.04, 2.18, 69.18, 36.76, 67.77, 40.94, 37.88, 6.44, 24.03, 31.21, 44.02, 50.53, 1.29, 78.89, 78.18, 41.64, 13.12, 60.07, 37.38, 7.18, 38.98, 75.08, 17.65, 11.13, 19.43, 44.85, 63.36, 68.78, 91.34, 71.68, 90.47, 82.64, 15.24, 24.96, 14.07, 1.79, 35.54, 62.56, 76.10, 7.22, 96.00, 49.04, 93.72, 41.35, 6.35, 31.58,

14.91, 46.10, 48.83, 25.15, 15.98, 18.06, 54.47, 88.50, 87.06, 16.53, 47.53, 46.51, 68.91, 28.35, 68.89, 93.41, 21.07, 25.42, 93.02, 94.64, 93.94, 9.75, 89.52, 1.20, 34.41, 65.48, 58.09, 32.27, 55.53, 22.79, 5.41, 87.13, 52.83, 14.59, 17.07, 29.37, 94.04, 24.53, 62.59, 58.07, 83.71, 86.55, 23.05, 34.52, 35.22, 50.84, 71.60, 27.66, 5.29, 72.83, 2.01, 5.79, 72.59, 94.20, 1.20, 91.42, 34.59, 68.72, 6.51, 29.91, 40.08, 45.36, 16.20, 1.91, 22.10, 86.63, 44.83, 6.13, 93.60, 73.23.

a) Write R code to construct frequency distribution table of width 15.

b) And also find relative frequency distribution and cumulative frequency distribution for the male students.

c) Represent the above (a) and (b) in column format.

R code with Output:

```
y=c(21.08, 13.76, 75.89, 68.41, 51.36, 21.58, 13.85, 82.75, 71.88, 57.93, 6.68, 24.99, 10.24,
24.60, 95.14, 99.86, 42.72, 3.66, 66.89, 96.60, 30.08, 49.50, 16.84, 89.14, 14.97, 66.86, 38.32,
71.90, 9.50, 66.92, 58.83, 38.06, 79.95, 95.30, 97.51, 49.67, 50.54, 93.64, 74.30, 76.82, 63.32,
36.54, 90.87, 72.37, 61.98, 89.44, 98.19, 27.07, 63.97, 55.35, 92.40, 32.97, 61.37, 34.40, 42.48,
77.67, 99.31, 54.64, 69.46, 49.88, 86.76, 31.95, 4.55, 36.77, 36.91, 56.45, 54.44, 76.53, 15.97,
18.22, 41.34, 50.17, 18.82, 96.60, 93.58, 8.85, 30.27, 44.18, 26.85, 54.82, 64.24, 46.20, 3.00,
96.14, 10.01, 75.64, 93.18, 93.32, 15.88, 84.66, 37.91, 97.11, 66.32, 87.91, 85.58, 64.32, 16.11,
37.22, 42.69, 89.64, 89.96, 48.29, 27.71, 89.71, 23.08, 53.02, 4.22, 18.84, 17.41, 73.05, 33.26,
71.83, 8.28, 69.65, 51.27, 75.61, 63.67, 8.24, 1.15, 83.05, 23.08, 26.37, 19.74, 34.79, 53.58,
66.88, 99.58, 63.30, 60.82, 43.71, 39.76, 54.97, 13.55, 81.42, 26.04, 2.18, 69.18, 36.76, 67.77,
40.94, 37.88, 6.44, 24.03, 31.21, 44.02, 50.53, 1.29, 78.89, 78.18, 41.64, 13.12, 60.07, 37.38,
7.18, 38.98, 75.08, 17.65, 11.13, 19.43, 44.85, 63.36, 68.78, 91.34, 71.68, 90.47, 82.64, 15.24,
24.96, 14.07, 1.79, 35.54, 62.56, 76.10, 7.22, 96.00, 49.04, 93.72, 41.35, 6.35, 31.58, 14.91,
46.10, 48.83, 25.15, 15.98, 18.06, 54.47, 88.50, 87.06, 16.53, 47.53, 46.51, 68.91, 28.35, 68.89,
93.41, 21.07, 25.42, 93.02, 94.64, 93.94, 9.75, 89.52, 1.20, 34.41, 65.48, 58.09, 32.27, 55.53,
22.79, 5.41, 87.13, 52.83, 14.59, 17.07, 29.37, 94.04, 24.53, 62.59, 58.07, 83.71, 86.55, 23.05,
34.52, 35.22, 50.84, 71.60, 27.66, 5.29, 72.83, 2.01, 5.79, 72.59, 94.20, 1.20, 91.42, 34.59,
68.72, 6.51, 29.91, 40.08, 45.36, 16.20, 1.91, 22.10, 86.63, 44.83, 6.13, 93.60, 73.23)
y
range(y)
breaks=seq(0,100,by=20)
C_I=cut(x, breaks, right = FALSE)
Freq=table(C_I)
Freq
Rel_Freq=table(C_I)/length(C_I)
Cum_Freq=cumsum(Freq)
Rel_Freq
Cum_Freq
cbind(Freq, Rel_Freq, Cum_Freq)
```

```
range(y)
[1]  1.15 99.86
> breaks=seq(0,100,by=20)
> C_I=cut(y, breaks, right = FALSE)
> Freq=table(C_I)
> Freq
C_I
   [0,20)  [20,40)  [40,60)  [60,80) [80,100)
       55       51       45       50       49
```

```
> Rel_Freq=table(C_I)/length(C_I)
> Cum_Freq=cumsum(Freq)
> Rel_Freq
C_I
   [0,20)  [20,40)  [40,60)  [60,80) [80,100)
    0.220    0.204    0.180    0.200    0.196
> Cum_Freq
   [0,20)  [20,40)  [40,60)  [60,80) [80,100)
       55      106      151      201      250
> cbind(Freq, Rel_Freq, Cum_Freq)
         Freq Rel_Freq Cum_Freq
[0,20)     55    0.220       55
[20,40)    51    0.204      106
[40,60)    45    0.180      151
[60,80)    50    0.200      201
[80,100)   49    0.196      250
```

Viva Questions

1. Write R Code to represent the values in column format.

   cbind()

2. Define range

   The difference between maximum and minimum value

3. Write R code to find the relative frequencies

   table(datavector)/length(datavector)

3. Acceptance/Rejection Sampling in R

Algorithm

1. Find a r.v. Y with density "g" satisfying $f(t)/g(t) \leq c$, for all t such that $f(t) > 0$.
2. Generate a random y from the dist. with density g
3. Generate a random u from the U (0, 1) dist.
4. If $u < f(y)/ (cg(y))$ accept y and deliver x = y; otherwise reject y and repeat step 2-4.

R code:

1. U[0,1] vs U[0,1]

```
>install.packages("AR")
>library(AR)
> simulation = AR.Sim( n = 10,
 f_X = function(y){dunif(y,min=0,max=1)},
 Y.dist = "unif",
 Y.dist.par = c(0,1),
 Rej.Num = TRUE,
 Rej.Rate = TRUE,
 Acc.Rate = FALSE)
>simulation
```

2. Uniform (0,1) VS Standard Normal (0,1)

```
simulation = AR.Sim( n = 10,
f_X = function(y){dunif(y,min=0,max=1)},
Y.dist = "norm",
Y.dist.par = c(0,1),
Rej.Num = TRUE,
Rej.Rate = TRUE,
Acc.Rate = FALSE)
>simulation
```

3. Uniform (0,1) VS Normal (mean,var)

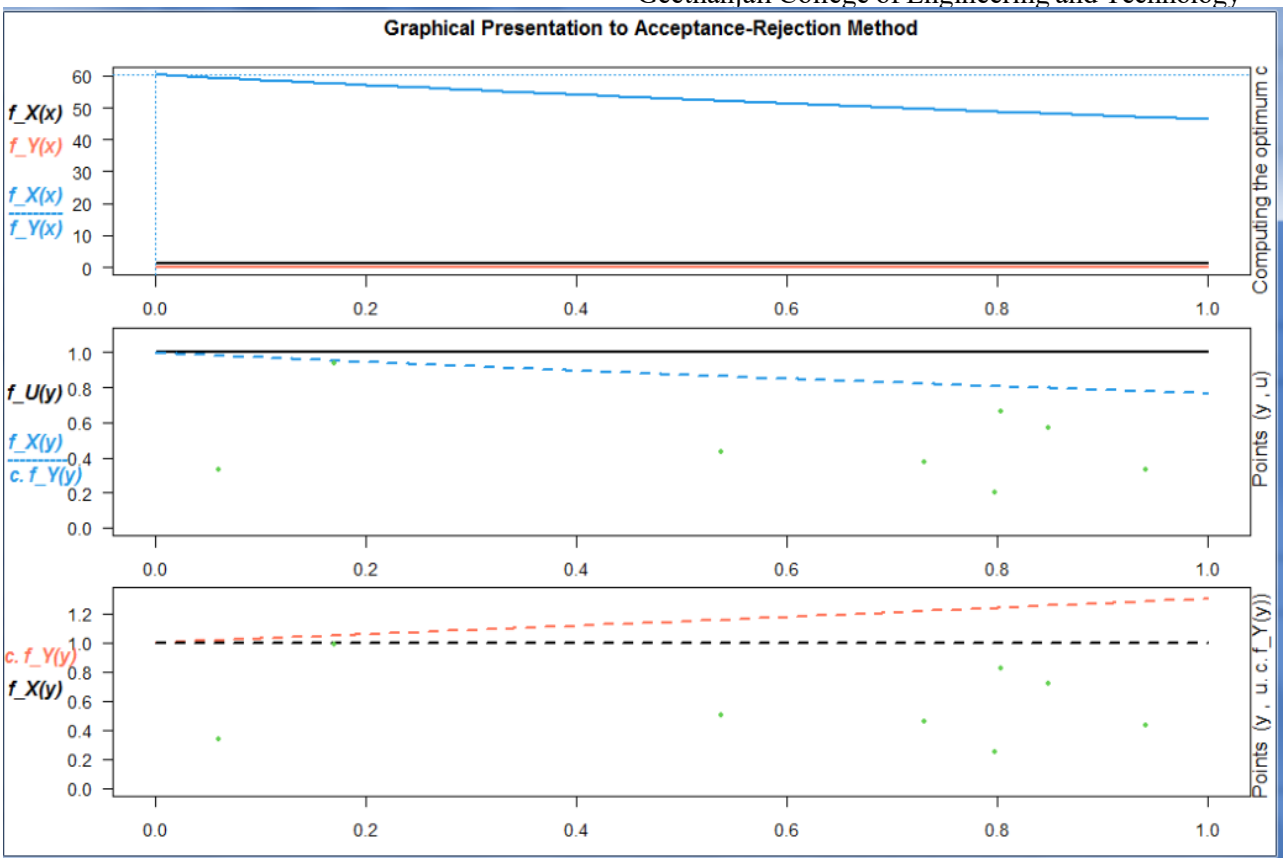
```
simulation = AR.Sim( n = 10,
f_X = function(y){dunif(y,min=0,max=1)},
Y.dist = "norm",
Y.dist.par = c(mean,s.d),
Rej.Num = TRUE,
Rej.Rate = TRUE,
Acc.Rate = FALSE)
>simulation
```

1. **Example:** Write R Program to simulate the data for n=8 using Acceptance-Rejection Algorithm for Uniform distribution in the interval (0,1) and Normal Distribution with parameters mean 10 and variance 36 and Comment on your result.

R Code with Output:

```
install.packages("AR")
library(AR)
simulation = AR.Sim( n = 8,
                     f_X = function(y){dunif(y,min=0,max=1)},
                     Y.dist = "norm",
                     Y.dist.par = c(10,6),
                     Rej.Num = TRUE,
                     Rej.Rate = TRUE,
                     Acc.Rate = FALSE)
simulation
```

```
> simulation = AR.Sim( n = 8,
+                      f_X = function(y){dunif(y,min=0,max=1)},
+                      Y.dist = "norm",
+                      Y.dist.par = c(10,6),
+                      Rej.Num = TRUE,
+                      Rej.Rate = TRUE,
+                      Acc.Rate = FALSE)
Optimal c = 60.314
The numbers of Rejections = 201
Ratio of Rejections = 0.962
> simulation
[1] 0.73086822 0.80376564 0.79708956 0.94135794 0.84846397 0.53779598 0.06042864 0.17044630
```



Inference: In this case, the rejection ration is 0.96 is significantly higher when sampling from Uniform distribution within the interval 0,1.

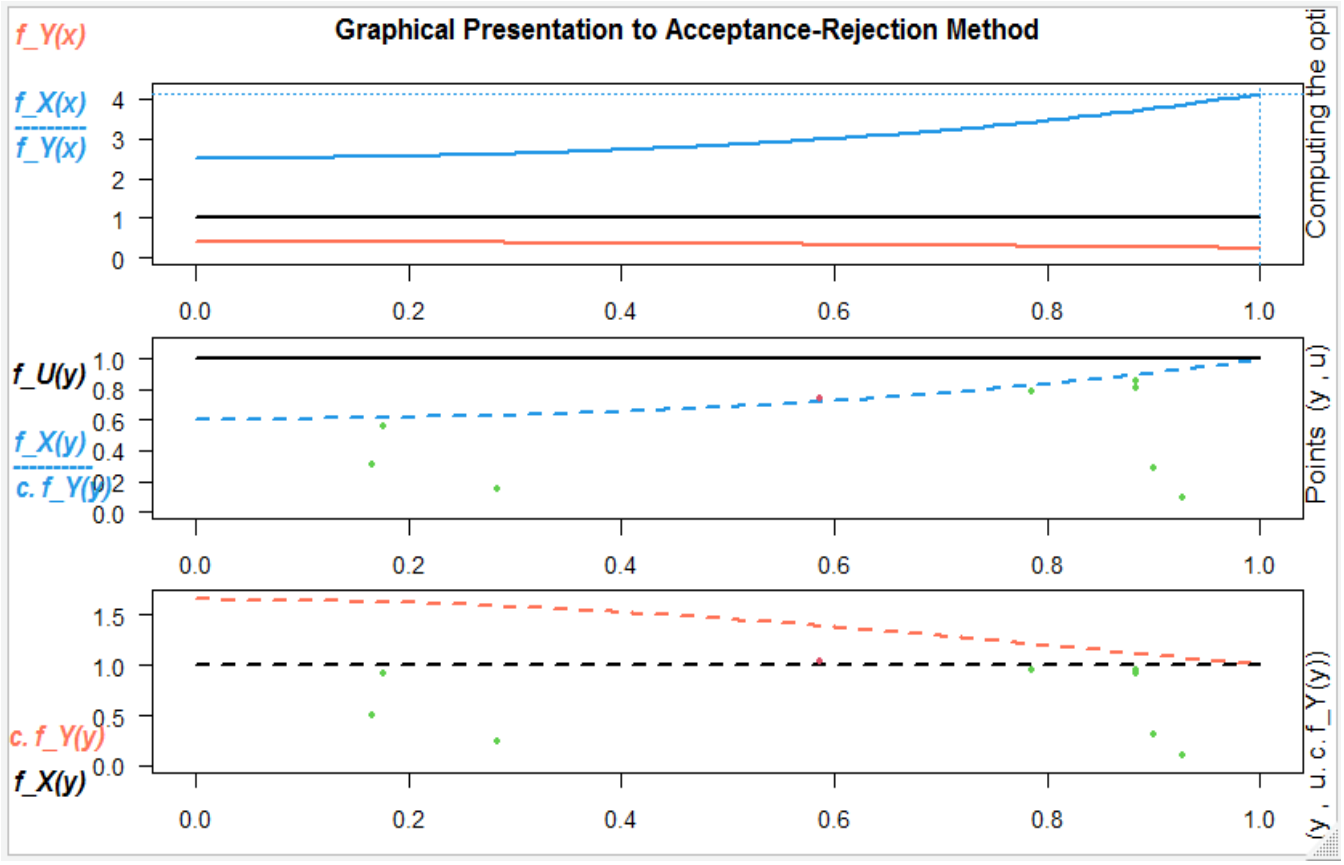
2. Example: Write R Program to generate 8 observations using Acceptance-Rejection Algorithm for Uniform distribution in the interval (0,1) and Standard Normal Distribution and Comment on your result.

R Code with Output:

```
simulation = AR.Sim( n = 8,
                     f_X = function(y){dunif(y,min=0,max=1)},
                     Y.dist = "norm",
                     Y.dist.par = c(0,1),
                     Rej.Num = TRUE,
                     Rej.Rate = TRUE,
                     Acc.Rate = FALSE)
```

simulation

```
> simulation = AR.Sim( n = 8,
+                      f_X = function(y){dunif(y,min=0,max=1)},
+                      Y.dist = "norm",
+                      Y.dist.par = c(0,1),
+                      Rej.Num = TRUE,
+                      Rej.Rate = TRUE,
+                      Acc.Rate = FALSE)
Optimal c = 4.132
The numbers of Rejections = 10
Ratio of Rejections = 0.556
> simulation
[1] 0.7860279 0.8839270 0.1767668 0.9269260 0.1661684 0.2832707 0.8832648
[8] 0.9009023
```



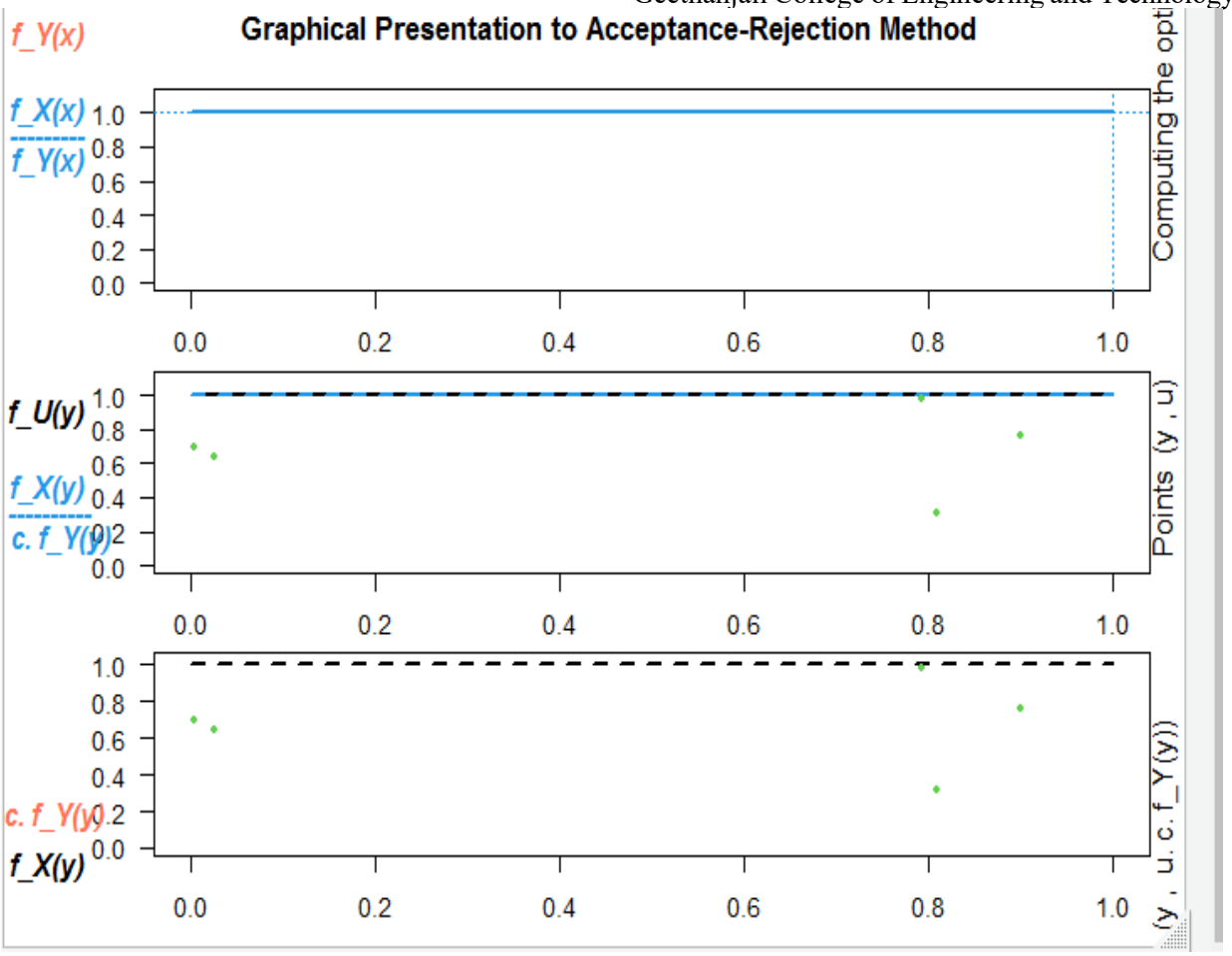
Inference: In this case, the rejection ration is 0.56 is not much significantly higher when sampling from Uniform distribution within the interval 0,1.

3. Example: Write R Program to generate 5 observations using Acceptance-Rejection Algorithm for Uniform distribution in the interval (0,1) and Uniform distribution in the interval (0,1) and Comment on your result.

R Code with Output:

```
simulation = AR.Sim( n = 5,
                     f_X = function(y){dunif(y,min=0,max=1)},
                     Y.dist = "unif",
                     Y.dist.par = c(0,1),
                     Rej.Num = TRUE,
                     Rej.Rate = TRUE,
                     Acc.Rate = FALSE)
```

```
> simulation = AR.Sim( n = 5,
+                      f_X = function(y){dunif(y,min=0,max=1)},
+                      Y.dist = "unif",
+                      Y.dist.par = c(0,1),
+                      Rej.Num = TRUE,
+                      Rej.Rate = TRUE,
+                      Acc.Rate = FALSE)
Optimal c = 1
The numbers of Rejections = 0
Ratio of Rejections = 0
> simulation
[1] 0.809260487 0.002256192 0.900072686 0.025223927 0.792977998
```



Inference: In this case, the rejection ration is 0. i.e generated random numbers are from same Uniform distribution. Hence all random numbers fall within the specified distribution

ViVa Questions

1. Which package is used to generate random numbers in 'R'language " AR"
2. The mean and Variance of Poisson distribution is ---------
   Mean=Variance=lamda
3. The basic technique used to generate observations from a distribution is
   Rejection sampling

4. Maximum likelihood and Method of moment's estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R

4.1 Maximum likelihood Estimation & Method of Moments estimation:

R Code:

> x=c(data vector)

>n=length(x) #No. of observations

>phat=mean(x)/n #MLE for Binomial distribution

where **n** represents no.of independent trails in Binomial distribution

>lambdahat=mean(x) #MLE for Poisson distribution

>muhat=mean(x) #MLE of Mean for Normal distribution

>sigmahat=((n - 1)/n)*var(x) #MLE of variance for Normal distribution

Note: If the sample data is not given then generate the sample using random numbers of a particular distribution and then estimate the corresponding parameters.

R Code to generate the random numbers from Binomial, Poisson and Normal distribution

rbinom(no of random numbers to be generate, n, p)

rpois(n,lambda)

rnorm(n, mean, sd)

1. Example: Sample (9,2,5,3,1,5,2,4,2,1,3,5,6,1,1) is drawn from Binomial distribution. Use R program to find the maximum likelihood estimate of the parameter p and estimate the mean and variance.

R Code with Output:

```
> x=c(9,2,5,3,1,5,2,4,2,1,3,5,6,1,1)
> phat=mean(x)/length(x)
> phat
[1] 0.2222222
> meanhat=length(x)*phat
> meanhat
[1] 3.333333
> varhat=length(x)*phat*(1-phat)
> varhat
[1] 2.592593
```

2. Example: Let X follows Poisson distribution with parameter 12. Use R program to generate 10 samples from Poisson distribution and hence find the maximum likelihood estimate of the parameter and estimate the mean and variance.

R Code with Output:

```
> x=rpois(10,12)
> lambdahat=mean(x)
> lambdahat
[1] 9.3
> meanhat=lambdahat
> meanhat
[1] 9.3
> varhat=lambdahat
> varhat
[1] 9.3
```

3. Example: Sample (15, 9, 3, 7, 15, 8, 10, 3, 12, 5, 15, 8, 10, 3, 15, 8, 10, 3, 15, 8, 10, 3, 15, 7, 14, 10, 10, 14, 10, 9, 3, 3, 10, 15, 8, 10, 3, 15, 8, 10, 3, 12, 5, 15, 8, 10, 3, 15, 8, 10, 3) is drawn from Normal distribution. Use R program to estimate the mean and variance using method of moments.

R Code with Output:

```
z=c(15,9,3,7,15,8,10,3,12,5,15,
    8,10,3,15,8,10,3,15,8,10,3,
    15,7,14,10,10,14,10,9,3,3,10,
    15,8,10,3,15,8,10,3,12,5,15,
    8,10,3,15,8,10,3)
muhat=mean(z)
muhat
sigmahat=((51 - 1)/51)*var(z)
sigmahat
```

```
muhat=mean(z)
> muhat
[1] 9.039216
> sigmahat=((51 - 1)/51)*var(z)
> sigmahat
[1] 17.3318
```

4. Example: Sample (9, 3, 7, 12, 5, 15, 15, 8, 10, 3, 15, 7, 14, 10, 10, 14, 10, 9, 3, 3, 10) is drawn from Poisson distribution. Use R program to estimate the parameter, mean and variance of Poisson using Method of Moments.

R Code with Output:

```
Y=c(9,3,7,12,5,15,15,8,10,3,15,7,14,10,10,14,10,9,3,3,10)
lambdahat=mean(Y)

lambdahat

[1] 9.142857
```

```
meanhat=lambdahat
meanhat
[1] 9.142857
varhat=lambdahat
varhat

[1] 9.142857
```

4.2 Testing of Hypothesis (Large Samples)

> **Compute one sample proportion of z-test (Test for single proportion)**

R Code:

prop.test(x, n, p, alternative = "two.sided")

prop.test(x, n, p, alternative = "less")

prop.test(x, n,p, alternative = "greater")

x: the number of successes

n: the total number of trials

p: the probability to test against.

1. Example: A manufacture of spark plugs claims that 3% of items supplied by him defective. A random sample of 500 plugs is found to have 20 defectives items. test the claim of manufacture.

R Code with Output:

prop.test(20,500,0.03,alternative="two.sided")

```
prop.test(20,500,0.03,alternative="two.sided")

        1-sample proportions test with continuity correction

data:  20 out of 500, null probability 0.03
X-squared = 1.3918, df = 1, p-value = 0.2381
alternative hypothesis: true p is not equal to 0.03
95 percent confidence interval:
 0.02524906 0.06216390
sample estimates:
   p
0.04
```

From the output we observe that:

Null Hypothesis (H_0): p = 0.03

Alternative Hypothesis (H_1): p ≠ 0.03

And the p value is 0.2381 > 0.05 level of significance

Hence We Accept Null hypothesis

95 % Confidence interval is (0.0252, 0.0621)

Phat = 0.04

2. Example: We have a population of mice containing half male and have female (p = 0.5 = 50%). Some of these mice (n = 160) have developed a spontaneous cancer, including 95 male and 65 female. Test whether the cancer affects more male than female?

R Code with Output:

prop.test(95,160,0.5,alternative="greater")

```
prop.test(95,160,0.5,alternative="greater")

        1-sample proportions test with continuity correction

data:  95 out of 160, null probability 0.5
X-squared = 5.2562, df = 1, p-value = 0.01093
alternative hypothesis: true p is greater than 0.5
95 percent confidence interval:
 0.5256923 1.0000000
sample estimates:
      p
0.59375
```

From the output we observe that:

Null Hypothesis (H_0): p = 0.5

Alternative Hypothesis (H_1): p > 0.5

And the p value is 0.01093 < 0.05 level of significance (los)

Hence We Reject Null hypothesis

95 % Confidence interval is (0.525, 1.0)

Phat = 0.593

> Compute two-proportions of z-test (Test for difference of proportions)

R Code:

prop.test(x=c(…..), n=c(……), p = NULL, alternative = " .....")

1. Example: In a large city A, consists of 900 school children, 20 had defective eyesight. In other large city B of a random sample 600 children, 15 had the same defect. Is this difference between the two proportions significant?

R Code with Output:

prop.test(x=c(20,15),n=c(900,600),p=NULL,alternative="two.sided")

```
prop.test(x=c(20,15),n=c(900,600),p=NULL,alternative="two.sided")

2-sample test for equality of proportions with continuity correction

data:  c(20, 15) out of c(900, 600)
X-squared = 0.030473, df = 1, p-value = 0.8614
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.01994014  0.01438458
sample estimates:
    prop 1     prop 2
0.02222222 0.02500000
```

From the output we observe that:

Null Hypothesis (H_0): p1=p2

Alternative Hypothesis (H_1): p1≠p2

And the p value is 0.8614 > 0.05 level of significance

Hence We Accept Null hypothesis

95 % Confidence interval is (-0.019, 0.014)

P1hat = 0.0222 P2hat=0.025

2. Example: In a company 500 people are having lung cancer and 500 people are healthy individuals. The number of smokers in each group is 490 and 400. Test whether the proportions of smokers are the same in the two groups of individuals?

R Code with Output:

prop.test(x=c(490,400),n=c(500,500),p=NULL,alternative="two.sided")

```
prop.test(x=c(490,400),n=c(500,500),p=NULL,alternative="two.sided")

        2-sample test for equality of proportions with continuity correction

data:  c(490, 400) out of c(500, 500)
X-squared = 80.909, df = 1, p-value < 2.2e-16
alternative hypothesis: two.sided
95 percent confidence interval:
 0.1408536 0.2191464
sample estimates:
prop 1 prop 2
  0.98   0.80
```

From the output we observe that:

Null Hypothesis (H_0): p1=p2

Alternative Hypothesis (H_1): p1≠p2

And the p value is 2.2e-16 < 0.05 level of significance

Hence We Reject Null hypothesis

95 % Confidence interval is (0.1408, 0.2191)

P1hat = 0.98 P2hat=0.80

C. One sample Z test in R (Test for single mean)

R Code:

install.packages("BSDA") #BSDA: Basic Statistics and Data Analysis
library(BSDA)
z.test(datavector, mu=population mean, sigma.x=……., alternative= "……")

1. Example: The price of a popular tennis racket at a national chain store is $179. Portia bought ten of the same racket at an online auction site for the following prices:

155,179,175,170,175,173,172,171,170,169,168,164,165,162,163,172,176,174,168,165,170,172, 168,162,175,161,160,158,156,171,170. If the auction prices of rackets are normally distributed, Use R code to determine whether there is sufficient evidence in the sample, at the 5% level of

significance, to conclude that the average price of the racket is less than $179 with standard deviation 6.1 if purchased at an online auction.

R Code with Output:

```
install.packages("BSDA")  #BSDA: Basic Statistics and Data Analysis
library(BSDA)
x=c(155,179,175,170,175,173,172,171,170,169,168,164,165,162,163,172,176,174,168,165,170,172,168,
   162,175,161,160,158,156,171,170)
z.test(x,mu=179,sigma.x=6.1,alternative="less")
```

```
X
 [1] 155 179 175 170 175 173 172 171 170 169 168 164 165 162 163 172 176 174
168 165 170
[22] 172 168 162 175 161 160 158 156 171 170
> z.test(X,mu=179,sigma.x=6.1,alternative="less")

        One-sample z-Test

data:  X
z = -10.011, p-value < 2.2e-16
alternative hypothesis: true mean is less than 179
sample estimates:
mean of x
 168.0323
```

From the output we observe that:

Null Hypothesis (H_0): $\mu=179$

Alternative Hypothesis (H_1): $\mu<179$

And z = -10.011; the p value is 2.2e-16 < 0.05 level of significance

Hence We Reject Null hypothesis

Meanhat=168.032

d.Two Sample Z-Test in R (Test for difference of means)

R Code:

z.test(datavector1,datavector2, alternative='……', mu=0, sigma.x= , sigma.y=)

1. Example: Suppose the IQ levels among individuals in two different cities are known to be normally distributed each with population standard deviations of 15 and 12 respectively. A scientist wants to test whether the mean IQ level between individuals in city A and city B are different, so she selects a simple random sample of 32 from city A and 36 samples from city B and records their IQ levels. **[Hint:** First generate random samples using rnorm]

R Code with Output:

```
cityA=rnorm(32,0,15)
cityB=rnorm(36,0,12)
z.test(cityA,cityB,mu=0,sigma.x=15,sigma.y=12,alternative="two.sided")
```

```
cityA=rnorm(32,0,15)
cityB=rnorm(36,0,12)
> z.test(cityA,cityB,mu=0,sigma.x=15,sigma.y=12,alternative="two.sided")

        Two-sample z-Test

data:  cityA and cityB
z = 0.47479, p-value = 0.6349
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -4.932750  8.086635
sample estimates:
mean of x mean of y
-2.393633 -3.970575
```

From the output we observe that:

Null Hypothesis (H_0): $\mu_1 = \mu_2$

Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$

And z = 0.4747; the p value is 0.6349 >0.05 level of significance

Hence We Accept Null hypothesis

95 % Confidence interval is (-4.9327, 8.0866)

Meanhat1= -2.3936 Meanhat2=-3.9705

Viva Questions:

1. Define Null Hypothesis and Alternative Hypothesis.

   Null Hypothesis (H_0): A null hypothesis is the hypothesis which asserts that there is no significant difference between the sample and the population parameter.

   Alternative Hypothesis (H_1): Any hypothesis which contradicts the Null hypothesis is called Alternative hypothesis.

2. Write the test statistic to test the difference of two proportions.

$$z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim (0,1)$$

3. Write the test-statistic to test for single mean in large samples.

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim (0,1)$$

5. Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R (Small Samples)

5.1 t test

i) t-test for single mean

R Code:

t.test(data vector, mu = population mean, alternative= "two.sided", conf.level=0.98)

t.test(data vector, mu = population mean, alternative= "less")

t.test(data vector, mu = population mean, alternative= "greater")

Note: Conf.level is an additional argument when one wants to construct 90% or 98% or 99%, we need to write. Otherwise by default it will construct 95% confidence interval without writing this additional argument.

1. Example: A random sample of 10 envelopes is taken from the letter box of a post office and their weights in grams are found to be 13.1, 14.5, 12.1, 11.9, 12.4, 12.3, 11.5, 11.6, 12.1 and 12.4. Does this sample indicate at 1% level that the average weight of envelopes received at their post office is 12.35 grams? Also construct 99% confidence interval for population mean.

R Code with Output:

```
A=c(13.1,14.5,12.1,11.9,12.4,12.3,11.5,11.6,12.1,12.4)

t.test(A,mu=12.35,alternative="two.sided", conf.level = 0.99)
```

```
A=c(13.1,14.5,12.1,11.9,12.4,12.3,11.5,11.6,12.1,12.4)
> t.test(A,mu=12.35,alternative="two.sided", conf.level = 0.99)

        One Sample t-test

data:  A
t = 0.14564, df = 9, p-value = 0.8874
alternative hypothesis: true mean is not equal to 12.35
99 percent confidence interval:
 11.49743 13.28257
sample estimates:
mean of x
    12.39
```

From the output we observe that:

Null Hypothesis (H_0): $\mu_1 = 12.35$

Alternative Hypothesis (H_1): $\mu_1 \neq 12.35$

And t = 0.14564; the p value is 0.8874> 0.01 level of significance
Hence We do not Reject Null hypothesis.

99 % Confidence interval is

```
(11.4974, 13.2825)
mean of x
    12.39
```

2. Example: Test the hypothesis that the average content of containers of a particular lubricant is 10 liters if the content of a random sample of 10 containers are:

10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, 9.8

Use 0.01 l.o.s and assume that the distribution of contents is normal. Also construct 95% confidence interval for population mean.

R Code with Output:

x=c(10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, 9.8)

t.test(x,mu=10,alternative="two.sided")

```
x=c(10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, 9.8)
> t.test(x,mu=10,alternative="two.sided")

        One Sample t-test

data:  x
t = 0.77174, df = 9, p-value = 0.46
alternative hypothesis: true mean is not equal to 10
95 percent confidence interval:
  9.884126 10.235874
sample estimates:
mean of x
    10.06
```

From the output we observe that:

Null Hypothesis (H_0): $\mu_1 = 10$

Alternative Hypothesis (H_1): $\mu_1 \neq 10$

And t = 0.771; the p value is 0.46 > 0.05 level of significance

Hence We Accept Null hypothesis

95 % Confidence interval is (9.884, 10.235)

Meanhat= 10.06

ii) Two sample t test (Test for difference of means_independent samples)

R Code:

t.test(data vector 1, data vector 2, mu = 0, alternative= " ..... ")

1. **Example:** Two Horses A and B were tested according to the time (seconds) in a particular track with the following results.

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	-

Test whether the two horses have same running capacity? Also construct 95% confidence interval for population mean.

R Code with Output:

HorseA=c(28,30,32,33,33,29,34)

HorseB=c(29,30,30,24,27,29)

t.test(HorseA,HorseB,mu=0,alternative="two.sided")

```
t.test(HorseA,HorseB,mu=0,alternative="two.sided")

        Welch Two Sample t-test

data:  HorseA and HorseB
t = 2.4335, df = 10.652, p-value = 0.03386
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.2867853 5.9513100
sample estimates:
mean of x mean of y
 31.28571  28.16667
```

From the output we observe that:

Null Hypothesis (H_0): $\mu_1 = \mu_2$

Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$

And t = 2.4335; the p value is 0.033 < 0.05 level of significance

Hence, We Reject Null hypothesis

95 % Confidence interval is (0.286, 5.951)

Meanhatx= 31.285 Meanhaty= 28.166

iii) Paired t test (Test for difference of means_dependent samples)

R Code:

t.test(after, before, paired = TRUE, alternative = ".....")

1. Example: Memory capacity of 10 students was tested before and after training. Test whether the training was effective or not from the following scores. Also construct 99% confidence interval for population mean.

Before training	12	14	11	8	7	10	3	0	5	6
After training	15	16	10	7	5	12	10	2	3	8

R Code with Output:

before=c(12,14,11,8,7,10,3,0,5,6)
after=c(15,16,10,7,5,12,10,2,3,8)
t.test(before,after,paired=TRUE,alternative="two.sided")

```
Paired t-test

data:  before and after
t = -1.3646, df = 9, p-value = 0.2055
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -3.1893268  0.7893268
sample estimates:
mean difference
           -1.2
```

From the output we observe that:

Null Hypothesis (H_0): $\mu_1 = \mu_2$

Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$

And t = -1.3646; the p value is 0.2055 > 0.05 level of significance

Hence, We Accept Null hypothesis

95 % Confidence interval is (-3.189, 0.789)

Meandifference = -1.2

5.2 Chi-square test

R Code:

x=matrix(c(data),ncol= ..... ,nrow=……,byrow=TRUE)
or
x =data.frame(col1 = c (…..), col2 = c(…..), ... **)**
x
chisq.test(x)

1. Example: 1000 students at college level were graded according to their I.Q and the economic conditions of their home. Write R code to test whether there is any association between condition at home and I.Q.

	I.Q		
Economic	High	Low	Total
Rich	460	140	600
Poor	240	160	400
Total	700	300	1000

R Code with Output:

x=matrix(c(460,140,240,160),ncol=2,nrow=2,byrow=TRUE)
x
chisq.test(x)

```
x=matrix(c(460,140,240,160),ncol=2,nrow=2,byrow=TRUE)
> x
     [,1] [,2]
[1,]  460  140
[2,]  240  160
> chisq.test(x)

        Pearson's Chi-squared test with Yates' continuity correction

data:  x
X-squared = 30.957, df = 1, p-value = 2.638e-08
```

From the output we observe that:

Null Hypothesis (H_0): Two attributes are independent

Alternative Hypothesis (H_1): Two attributes are not independent

And chi-square = 30.957; the p value is 0.0000 < 0.05 level of significance

Hence, We Reject Null hypothesis

Example: In an industry, 1000 workers, employed for a specific job, were classified according to their performance and training received to test independence of a specific training and performance. The data is summarized as follows.

	Performance	
Training	Good	Not Good
Trained	230	148
Untrained	151	471

Write R code to make use of Chi-square test for independence at 5 % level of significance and write your conclusion.

R Code with Output:

```
x=matrix(c(230,148,151,471),ncol=2,nrow=2,byrow=TRUE)
x
x
     [,1] [,2]
[1,]  230  148
[2,]  151  471

chisq.test(x)

chisq.test(x)

        Pearson's Chi-squared test with Yates' continuity correction

data:  x
X-squared = 131.78, df = 1, p-value < 2.2e-16
```

From the output we observe that:

Null Hypothesis (H_0): Two attributes specific training and performance are independent

Alternative Hypothesis (H_1): Two attributes specific training and performance are not independent

And chi-square = 131.78; the p value is 2.2e-16< 0.05 level of significance
Hence, We Reject Null hypothesis.

5.3 F-test. (Test for equality of variances)

R Code:

var.test(sample1,sample2,alternative="….")

Note: While calculating F ratio, F value should be greater than 1. If not interchange the samples.

1. Example: Two Horses A and B were tested according to the time (seconds) in a particular track with the following results.

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	-

Test at 5% level whether the populations have the same variance.

R Code with Output:

```
x=c(28,30,32,33,33,29,34)
y=c(29,30,30,24,27,29)
var.test(y,x,alternative="two.sided")
var.test(y,x,alternative="two.sided")

        F test to compare two variances

data:  y and x
F = 1.0245, num df = 5, denom df = 6, p-value = 0.9573
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1711122 7.1489727
sample estimates:
ratio of variances
          1.024545
```

From the output we observe that:

Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$

Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$

And F = 1.0245; the p value is 0.9573 > 0.05 level of significance

Hence, We Accept Null hypothesis

95 % Confidence interval is (0.1711, 7.148)

2. Example: The following are the values in thousands of an inch obtained by 2 engineers in 8 successive measurements with the same micrometer:

Engineer A: 503,505,497,505,419,493,510,501

Engineer B: 502,497,492,490,495,497,596,498

Is one engineer more consistent than the other?

R Code with Output:

```
x=c(503,505,497,505,419,493,510,501)
y=c(502,497,492,490,495,497,596,498)
var.test(y,x,alternative="two.sided")
F test to compare two variances

data:  y and x
F = 1.4265, num df = 7, denom df = 7, p-value = 0.651
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.2855932 7.1252993
sample estimates:
ratio of variances
          1.426512
```

From the output we observe that:

Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$

Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$

And F = 1.4265; the p value is 0.651 > 0.05 level of significance

Hence, We Accept Null hypothesis

95 % Confidence interval is (0.285, 7.125)

6. Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi collinearity in R

6.1 Fitting of Multiple Linear Regression

R Code:

```
MLR<- lm(Dependent variable ~ sum of independent variables)
```

- **Testing MLR**

R Code:

```
summary(MLR)
```

6.2. Outlier analysis

R Code:

```
plot(MLR)
```

6.3. Residual analysis

R Code

```
Residuals=resid(MLR)
Predicted=predict(MLR)
Residuals
Predicted
plot(Residuals,Predicted)
```

6.4. Test for normality

R Code:

```
plot(MLR,2)
```

2 indicates second plot i.e Normal Q-Q plot

```
shapiro.test(Residuals)
```

to test the Residuals follows normality or not

6.5. Multi Collinearity

R Code:

```
install.packages("caTools")
                                    # For Linear regression
install.packages('car')
                                    # To check multicollinearity
install.packages("quantmod")
install.packages("MASS")
install.packages("corrplot")
                                    # plot correlation plot
library(caTools)
library(car)
library(quantmod)
library(MASS)
library(corrplot)
vif(MLR)    #Variance Inflation Factor (vif) specifies whether Multicollinearity present in the variables or not
```

if vif value is >10 then there exist Multicollinearity

1. Example:

The diameter of a tree at its breast height is influenced by many factors. The below table relate the diameter of a particular type of eucalyptus tree to its age, average rainfall at its site, site's elevation, and the wood's mean specific gravity.

S.No.	Age (Years)	Elevation (1000 ft)	Rainfall (inches)	Specific Gravity	Diameter at Breast Height (inches)
1	35	1.5	110	0.63	18.1
2	36	1.2	115	0.59	19.6
3	34	2.1	80	0.56	16.6
4	35	2.3	89	0.55	16.4
5	36	2.5	78	0.54	16.9
6	32	1.6	54	0.59	17.0
7	35	1.5	86	0.56	20.0
8	38	1.4	110	0.46	16.6
9	39	2.3	118	0.63	16.2
10	34	2.5	119	0.60	18.5
11	32	2.6	200	0.63	18.7
12	31	1.2	240	0.58	19.4
13	35	1.3	210	0.55	17.6
14	36	1.8	110	0.57	18.3
15	32	1.9	118	0.62	18.8

a) Fit a Multiple linear regression model and Test the Model
b) Identify the Outliers in the data
c) Check for Multicollinearity for the variables
d) Compute the Residuals
e) Test for Normality

a) R Code with Output:

```
Age=c(35,36,34,35,36,32,35,38,39,34,32,31,35,36,32)
Elevation=c(1.5,1.2,2.1,2.3,2.5,1.6,1.5,1.4,2.3,2.5,2.6,1.2,1.3,1.8,1.9)
Rainfall=c(110,115,80,89,78,54,86,110,118,119,200,240,210,110,118)
Diameter=c(18.1,19.6,16.6,16.4,16.9,17,20,16.6,16.2,18.5,18.7,19.4,17.6,18.3,18.8)
Gravity=c(0.63,.59,.56,.55,.54,.59,.56,.46,.63,.60,.63,.58,.55,.57,.62)
MLR=lm(Diameter~Age+Elevation+Rainfall+Gravity)
summary(MLR)
```

```
Call:
lm(formula = Diameter ~ Age + Elevation + Rainfall + Gravity)

Residuals:
    Min      1Q  Median      3Q     Max
-1.4357 -0.7493 -0.1480  0.5501  2.0489
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 19.546563   7.820128   2.500   0.0315 *
Age         -0.146790   0.150096  -0.978   0.3512
Elevation   -1.038764   0.644716  -1.611   0.1382
Rainfall     0.003418   0.006245   0.547   0.5961
Gravity      8.582746   7.548318   1.137   0.2820
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

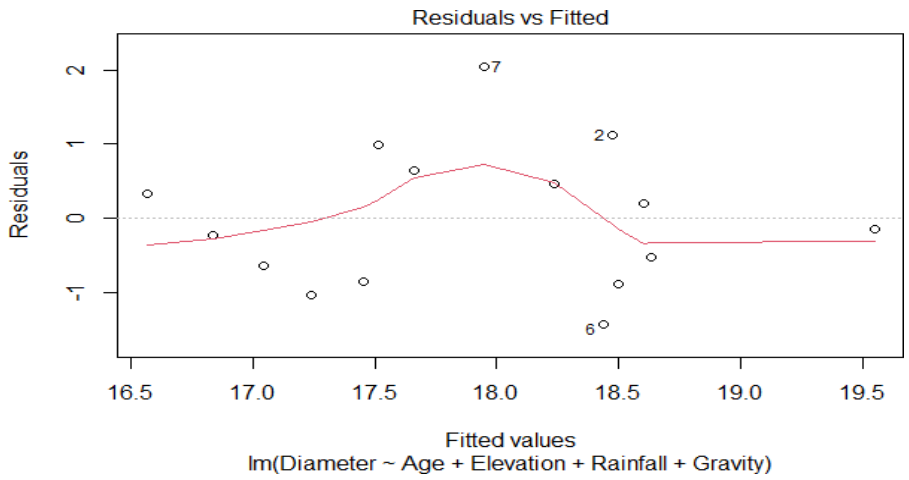
Residual standard error: 1.126 on 10 degrees of freedom
Multiple R-squared:  0.4233,  Adjusted R-squared:  0.1927
F-statistic: 1.835 on 4 and 10 DF,  p-value: 0.1987
```

Inference:

The Independent variables are not significant here because the corresponding p-values are greater than 0.05, so they cannot explain the variation in the dependent variable. i.e to explain variation in dependent variable (diameter of breast of tree)these independent variables are not sufficient, may be some other variables will be significant.

b) R Code with Output:

plot(MLR)



Inference: From the above graph 6,7,20 are outliers

c) R Code with Output:

```
install.packages("caTools") # For Linear regression
install.packages('car')  # To check multicollinearity
install.packages("quantmod")
install.packages("MASS")
install.packages("corrplot")  # plot correlation plot
library(caTools)
library(car)
library(quantmod)
library(MASS)
library(corrplot)
vif(MLR)
vif(MLR)
      Age Elevation  Rainfall    Gravity
 1.268037  1.160155  1.199486  1.272002
```

Inference:

No factor is having >10. Hence there is no multicollinearity

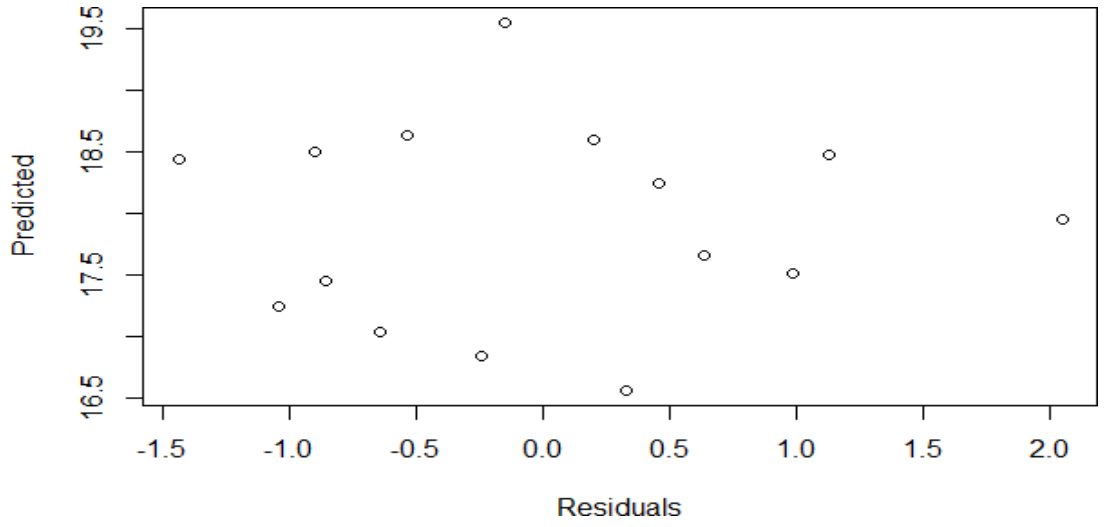
d) R Code with Output:

```
Residuals=resid(MLR)
Predicted=predict(MLR)
Residuals
Predicted
plot(Residuals,Predicted)
 Residuals=resid(MLR)
> Predicted=predict(MLR)
> Residuals
Residuals=resid(MLR)
> Predicted=predict(MLR)
> Residuals
      1          2          3          4          5          6          7
-0.5339163  1.1274623 -0.8541022 -0.6444976  0.3334760 -1.4356677  2.0489187
      8
-0.2383551

       9         10         11         12         13         14         15
-1.0430912  0.9847743  0.4606936 -0.1479675 -0.8968942  0.6394681  0.1996987
```

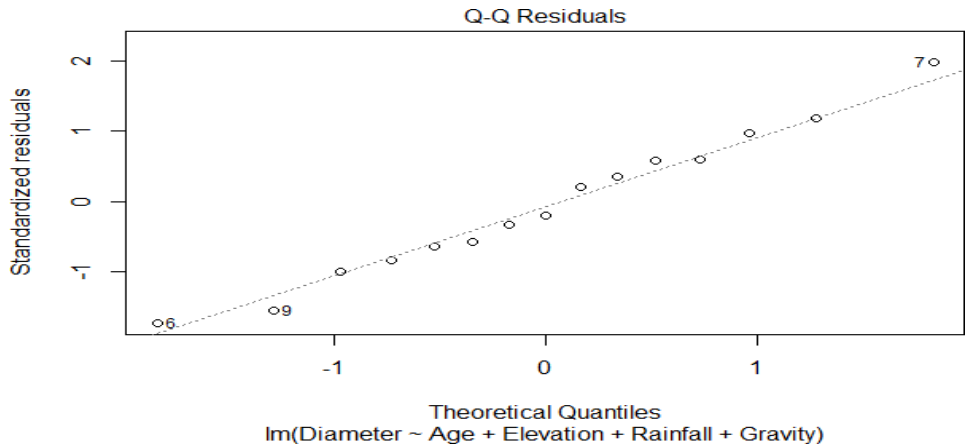
```
> Predicted
       1        2        3        4        5        6        7        8
  9
18.63392 18.47254 17.45410 17.04450 16.56652 18.43567 17.95108 16.83836 17.24
309
      10       11       12       13       14       15
17.51523 18.23931 19.54797 18.49689 17.66053 18.60030
Plot(Residuals,predicted)

plot(Residuals,Predicted)
```



e) R Code with Output:

plot(MLR,2)



shapiro.test(Residuals)

```
shapiro.test(Residuals)

        Shapiro-Wilk normality test

data:  Residuals
W = 0.97048, p-value = 0.8652
```

H_0 :The Residuals are normally distributed

H_1 :The Residuals are not Normally distributed

Inference: If p-value = 0.8652 > 0.05

Accept H_0.

2. Example: The data given below consists of 10 observations on the response (Y) effected by the independent variables X1, X2, X3, X4. Use R code:

a) Fit a Multiple linear regression model
b) Identify the Outliers in the data
c) Test for Normality

Y	21.0	22.0	21.0	23.0	21.2	23.2	21.5	22.0	22.3	22.5
X1	2.3	2.5	2.6	2.5	2.4	2.1	2.0	2.2	2.3	2.9
X2	3.1	3.2	3.3	3.4	3.5	3.6	3.8	3.9	3.2	3.3
X3	4.1	4.2	4.3	4.4	4.5	4.6	4.6	4.8	4.9	4.2
X4	5	6	7	8	9	1	5	4	2	6

a) R Code with Output:

Y=c(21.0,22.0,21.0,23.0,21.2,23.2,21.5,22.0,22.3,22.5)

X1=c(2.3,2.5,2.6,2.5,2.4,2.1,2.0,2.2,2.3,2.9)

X2=c(3.1,3.2,3.3,3.4,3.5,3.6,3.8,3.9,3.2,3.3)

X3=c(4.1,4.2,4.3,4.4,4.5,4.6,4.6,4.8,4.9,4.2)

X4=c(5,6,7,8,9,1,5,4,2,6)

mlr=lm(Y~X1+X2+X3+X4)

summary(mlr)

```
 Call:
lm(formula = Y ~ X1 + X2 + X3 + X4)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  13.7806     8.9768   1.535    0.185
X1            1.7994     1.5884   1.133    0.309
X2            0.8885     1.5039   0.591    0.580
X3            0.4200     1.5674   0.268    0.799
X4           -0.1913     0.1514  -1.264    0.262

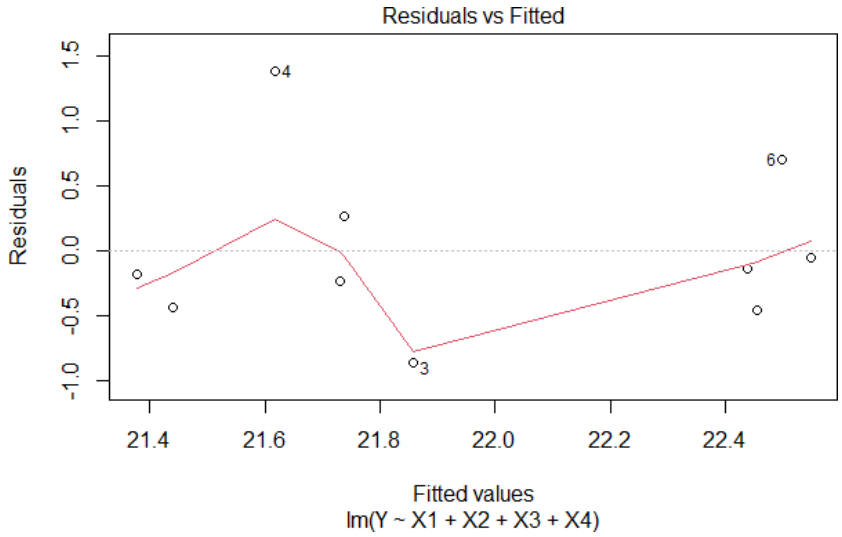
Residual standard error: 0.8618 on 5 degrees of freedom
Multiple R-squared:  0.344,    Adjusted R-squared:  -0.1808
F-statistic: 0.6555 on 4 and 5 DF,  p-value: 0.6483
```

Inference:

The Independent variables are not significant here because the corresponding p-values are greater than 0.05, so they cannot explain the variation in the dependent variable y.

b) R Code with Output:

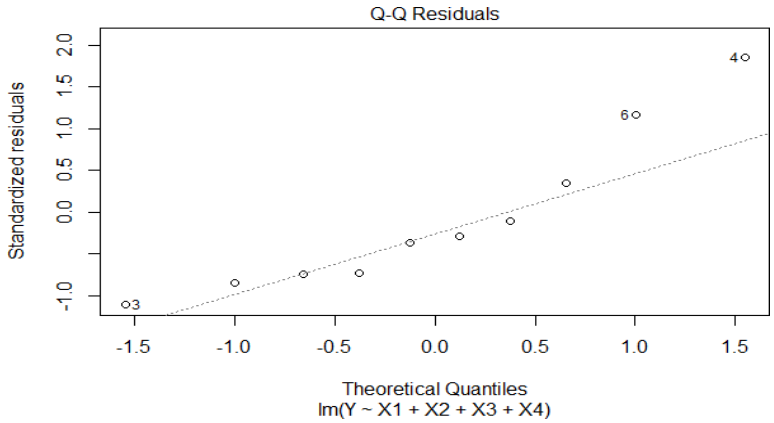
plot(mlr)



Inference: From the above graph 3,4,6 are outliers

C) R Code with Output:

plot(mlr,2)



shapiro.test(Residuals)

```
shapiro.test(Residuals)

        Shapiro-Wilk normality test

data: Residuals
W = 0.91186, p-value = 0.294
```

H_0 :The Residuals are normally distributed

H_1 :The Residuals are not Normally distributed

Inference: If p-value = 0.294> 0.05

Accept H_0.

3. Fit Multiple Linear regression to Marketing data which is available in "R".

a) Identify the Outliers in the data
b) Test for Normality

Note: This data can be downloaded using the package "datarium"

R Code with Output:

data("marketing", package = "datarium")

head(marketing, 4) #Displays first 4 values in the data

```
head(marketing, 4)
   youtube facebook   newspaper sales
1  276.12    45.36     83.04    26.52
2   53.40    47.16     54.12    12.48
3   20.64    55.08     83.16    11.16
4  181.80    49.56     70.20    22.20
```

tail(marketing) #Displays last 6 values in the data

```
     youtube facebook  newspaper  sales

195  179.64    42.72      7.20    20.76
196   45.84     4.44     16.56     9.12
197  113.04     5.88      9.72    11.64
198  212.40    11.16      7.68    15.36
199  340.32    50.40     79.44    30.60
200  278.52    10.32     10.44    16.08
```

View(marketing) #Displays the total data

model <- lm(sales ~ youtube + facebook + newspaper, data = marketing)

summary(model)

```
model <- lm(sales ~ youtube + facebook + newspaper, data = marketing)
> summary(model)

Call:
lm(formula = sales ~ youtube + facebook + newspaper, data = marketing)

Residuals:
     Min       1Q   Median       3Q      Max
-10.5932  -1.0690   0.2902   1.4272   3.3951

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.526667   0.374290   9.422   <2e-16 ***
youtube      0.045765   0.001395  32.809   <2e-16 ***
facebook     0.188530   0.008611  21.893   <2e-16 ***
newspaper   -0.001037   0.005871  -0.177     0.86
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.023 on 196 degrees of freedom
Multiple R-squared:  0.8972,  Adjusted R-squared:  0.8956
F-statistic: 570.3 on 3 and 196 DF,  p-value: < 2.2e-16
```

INFERENCE:

The p-values regarding intercept, YouTube and Facebook, are less than 0.05. So these variables are significant and also included in the model.

The model equation is sales =3.52667 +0.0457 *YouTube+0.1885*Facebook

Multiple R-square and Adjusted R-square are approximately equal for the above data. Adjusted R-square value is 0.89565.So the independent variables YouTube and Facebook explains 89 % variation in the dependent variable sales which plays major role.

- If one or more independent variables added to the model, then Multiple R Square value will increase whether it is significant or not and Adjusted R squared value either will decrease or constant in the case insignificant.
- If Adjusted R squared value increases then the added independent variable is significant and the variable will be included in the model.

F –test is used to test the overall significant of the model

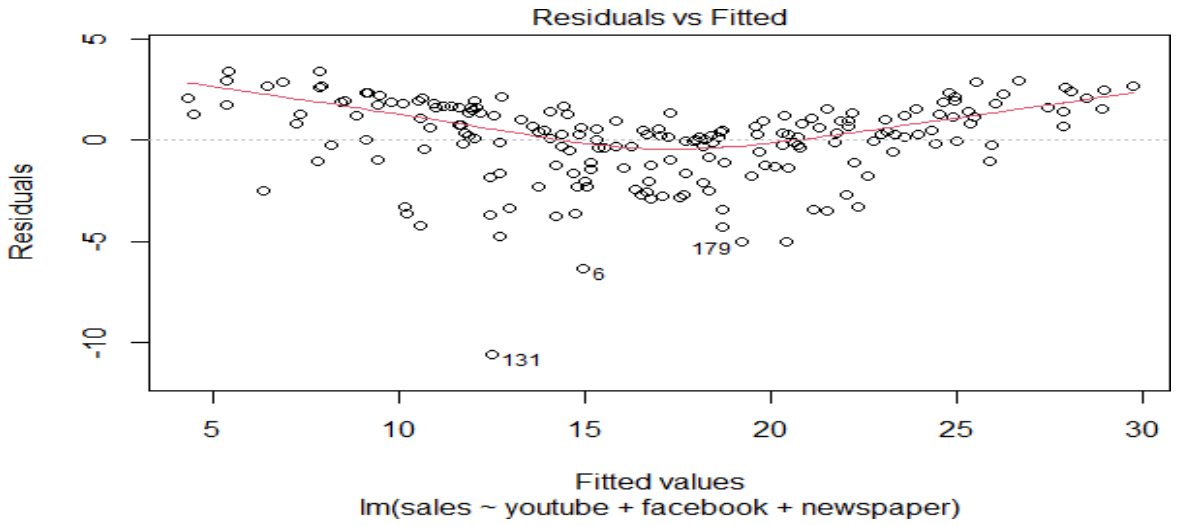
H_0: The model is not significant

H_1: The model is significant

The p-value for this f-statistic is less than 0.05 so we reject H_0.Therefore the model is significant to the data.

a) R Code with Output:

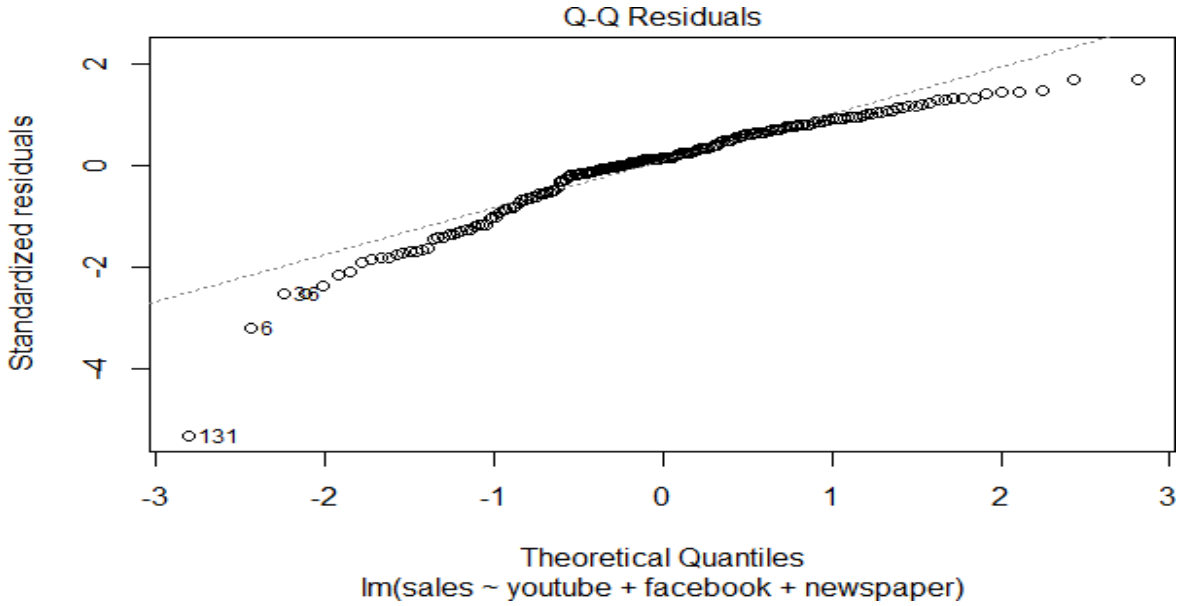
plot(model)



Inference: From the above graph 6,131,179 are outliers.

b) R Code with Output:

plot(model,2)




```
Shapiro-Wilk normality test

data:  Residuals
W = 0.97048, p-value = 0.8652
```

H_0 :The Residuals are normally distributed

H_1 :The Residuals are not Normally distributed

Inference: If p-value = 0.8652> 0.05

Accept H_0.

Viva Questions

1. Define Linear Regression.

   It is the mathematical average relationship between dependent and independent variables

2. Define Residual.

   Difference between actual response variable and predicted response variable

3. Define Multicollinearity and when Multicollinearity exist in linear regression.

   Multicollinearity occurs when two or more independent variables in a data frame have a high correlation with one another in a regression model.Multicollinearity exists whenever an independent variable is highly correlated with one or more of the other independent variables in a multiple regression equation..

7. Logistic Regression in R

R Code:

Logistic_Reg = glm(Dependent variable ~ sum of Independent variables, family="binomial")

summary(Logistic_Reg)

1. Example:

Consider data published on $n = 27$ leukemia patients. The data has a response variable of whether leukemia remission occurred (REMISS), which is given by a 1. The predictor variables are cellularity of the marrow clot section (CELL), smear differential percentage of blasts (SMEAR), percentage of absolute marrow leukemia cell infiltrate (INFIL), percentage labeling index of the bone marrow leukemia cells (LI), absolute number of blasts in the peripheral blood (BLAST), and the highest temperature prior to start of treatment (TEMP).

REMISS	CELL	SMEAR	INFIL	LI	BLAST	TEMP
1	0.8	0.83	0.66	1.9	1.1	1
1	0.9	0.36	0.32	1.4	0.74	0.99
0	0.8	0.88	0.7	0.8	0.18	0.98
0	1	0.87	0.87	0.7	1.05	0.99
1	0.9	0.75	0.68	1.3	0.52	0.98
0	1	0.65	0.65	0.6	0.52	0.98
1	0.95	0.97	0.92	1	1.23	0.99
0	0.95	0.87	0.83	1.9	1.35	1.02
0	1	0.45	0.45	0.8	0.32	1
0	0.95	0.36	0.34	0.5	0	1.04
0	0.85	0.39	0.33	0.7	0.28	0.99
0	0.7	0.76	0.53	1.2	0.15	0.98
0	0.8	0.46	0.37	0.4	0.38	1.01
0	0.2	0.39	0.08	0.8	0.11	0.99
0	1	0.9	0.9	1.1	1.04	0.99
1	1	0.84	0.84	1.9	2.06	1.02
0	0.65	0.42	0.27	0.5	0.11	1.01
0	1	0.75	0.75	1	1.32	1
0	0.5	0.44	0.22	0.6	0.11	0.99
1	1	0.63	0.63	1.1	1.07	0.99
0	1	0.33	0.33	0.4	0.18	1.01
0	0.9	0.93	0.84	0.6	1.59	1.02
1	1	0.58	0.58	1	0.53	1
0	0.95	0.32	0.3	1.6	0.89	0.99
1	1	0.6	0.6	1.7	0.96	0.99
1	1	0.69	0.69	0.9	0.4	0.99
0	1	0.73	0.73	0.7	0.4	0.99

R Code with Output:

```
Remiss=c(1,1,0,0,1,0,1,0,0,0,0,0,0,0,1,0,0,0,1,0,0,1,0,1,1,0)

cell=c(.8,.9,.8,1,.9,1,.95,.95,1,.95,.85,.7,.8,.2,1,1,.65,1,.5,1,1,.9,1,.95,1,1,1)

smear=c(.83,.36,.88,.87,.75,.65,.97,.87,.45,.36,.39,.76,.46,.39,.9,.84,.42,.75,.44,.63,.33,.93,.58,.3
2,.6,.69,.73)

infil=c(.66,.32,.7,.89,.68,.65,.92,.83,.45,.34,.33,.53,.37,.08,.9,.84,.27,.75,.22,.63,.33,.84,.58,.3,.6,.
69,.73)

l1=c(1.9,1.4,0.8,.7,1.3,.6,1,1.9,.8,.5,.7,1.2,.4,.8,1.1,1.9,.5,1,.6,1.1,.4,.6,1,1.6,1.7,0.9,0.7)

blast=c(1.1,.74,.18,1.05,0.52,0.52,1.23,1.35,0.32,0,0.28,0.15,0.38,0.11,1.04,2.06,0.11,1.32,0.11,
1.07,0.18,1.59,0.53,0.89,0.96,0.4,0.4)

temp=c(1,.99,.98,.99,.98,.99,.99,1.02,1,1.04,0.99,.98,1.01,.99,.99,1.02,1.01,1,.99,.99,1.01,1.02,1
,.99,.99,.99,.99)

logistic_Model=glm(Remiss~cell+smear+infil+l1+blast+temp,family="binomial")

logistic_Model
```

```
Call:  glm(formula = Remiss ~ cell + smear + infil + l1 + blast + temp,
    family = "binomial")

Coefficients:
(Intercept)         cell        smear        infil           l1        blast
     74.704       43.501       38.723      -39.902        4.354        0.359
       temp
   -122.924

Degrees of Freedom: 26 Total (i.e. Null);  20 Residual
Null Deviance:      34.37
Residual Deviance: 20.76        AIC: 34.76
```

Inference:

$$\hat{R}emiss = 74.704 + 43.501 * cell + 38.723 * smear - 39.902 * infil + 4.354 * l1 + 0.359 * blast - 122.924 * temp$$

summary(logistic_Model)

```
Call:
glm(formula = Remiss ~ cell + smear + infil + l1 + blast + temp,
    family = "binomial")

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-2.08053  -0.61978  -0.01379   0.75830   1.55672

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   74.704     74.685   1.000    0.317
cell          43.501     48.801   0.891    0.373
smear         38.723     56.436   0.686    0.493
infil        -39.902     59.808  -0.667    0.505
l1             4.354      2.725   1.598    0.110
blast          0.359      2.319   0.155    0.877
temp        -122.924     82.645  -1.487    0.137

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 34.372  on 26  degrees of freedom
Residual deviance: 20.761  on 20  degrees of freedom
AIC: 34.761

Number of Fisher Scoring iterations: 8
```

Inference:

All the p- values corresponding to coefficients are greater than 0.05,so these variables are not significant.

2. Fit Logistic Regression to Student admit data and write the appropriate results
Note: The data can be download using the following code

```
install.packages("packagename")

mydata <- read.csv("https://stats.idre.ucla.edu/stat/data/binary.csv")
```

R Code with Output:

```
install.packages("packagename")


mydata <- read.csv("https://stats.idre.ucla.edu/stat/data/binary.csv")
> View(mydata)
> summary(mydata)#Displays Min,1st quartile,median,mean,3rd quartile,max for all
                 variables
     admit            gre             gpa             rank
 Min.   :0.0000   Min.   :220.0   Min.   :2.260   Min.   :1.000
 1st Qu.:0.0000   1st Qu.:520.0   1st Qu.:3.130   1st Qu.:2.000
 Median :0.0000   Median :580.0   Median :3.395   Median :2.000
 Mean   :0.3175   Mean   :587.7   Mean   :3.390   Mean   :2.485
 3rd Qu.:1.0000   3rd Qu.:660.0   3rd Qu.:3.670   3rd Qu.:3.000
 Max.   :1.0000   Max.   :800.0   Max.   :4.000   Max.   :4.000
```

```
> str(mydata) #Displays the Structure of the data

'data.frame':  400 obs. of  4 variables:
 $ admit: int  0 1 1 1 0 1 1 0 1 0 ...
 $ gre  : int  380 660 800 640 520 760 560 400 540 700 ...
 $ gpa  : num  3.61 3.67 4 3.19 2.93 3 2.98 3.08 3.39 3.92 ...
 $ rank : int  3 3 1 4 4 2 1 2 3 2 ...
> mydata$rank <- factor(mydata$rank)
  mydata$admit<-factor(mydata$admit)
> mylogit <- glm(admit ~ gre + gpa + rank, data = mydata, family = "binomial"
)
> summary(mylogit)

Call:
glm(formula = admit ~ gre + gpa + rank, family = "binomial",
    data = mydata)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	-3.989979	1.139951	-3.500	0.000465	***
Gre	0.002264	0.001094	2.070	0.038465	*
Gpa	0.804038	0.331819	2.423	0.015388	*
rank2	-0.675443	0.316490	-2.134	0.032829	*
rank3	-1.340204	0.345306	-3.881	0.000104	***
rank4	-1.551464	0.417832	-3.713	0.000205	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 499.98 on 399 degrees of freedom
Residual deviance: 458.52 on 394 degrees of freedom
AIC: 470.52

Number of Fisher Scoring iterations: 4

INFERENCE

All the p-values corresponding to intercept,gre,gpa and rank in this model output are significant and also these are included in the model.

The logistic model equation is

Y= -3.989979 +0.002264 *gre + 0.804038 *gpa -0.675443 *rank2-1.340204 *rank3 -1.551464*rank4

The probability values will be calculated by using the following equation

Here $P = e^y / 1 + e^y$ **# this helps to convert continuous values in to categorical values by considering p>0.5 as 1 and P<0.5 as 0**

Note:Logistic Regression is a classification technique , it s called as logistic regression because, in out put it gives continuous vales.

Viva Questions:

1. What kind of Machine Learning is Logistic Regression?

   Supervised Machine Learning

2. Define Binary Response Variable?

   A binary response variable is a categorical variable that can only take one of two values

3. State any two differences of Linear Regression and Logistic Regression?

Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic Regression is used to predict the categorical dependent variable using a given set of independent variables.
Linear Regression is used for solving Regression problem.	Logistic regression is used for solving Classification problems.

8. Classification with SVM in R

R Code:

```
install.packages("e1071")
library(e1071)
x = matrix(rnorm(32), 16, 2)          #Generating the data
y = rep(c(-1, 1), c(8, 8))            #Assigning +1 and -1 to the above data
x[y == 1,] = x[y == 1,] + 1           #bias the x variable when y =1
dat = data.frame(x, y = as.factor(y)) #representing the above data in table form using dataframe
svm = svm(y ~ ., data = dat, kernel = "linear",cost=10)
summary(svm)
```

1. Example: Write R code to find the number of support vectors for classifying the data. Data as follows: 60 observations, in two columns and also get the SVM classification plot.

R Code with Output:

```
install.packages("e1071")
library(e1071)
x = matrix(rnorm(60), 30, 2)
y = rep(c(-1, 1), c(15, 15))
x[y == 1,] = x[y == 1,] + 1
data = data.frame(x, y = as.factor(y))
svm = svm(y ~ ., data = data, kernel = "linear",cost=10)
summary(svm)
```

```
Call:
svm(formula = y ~ ., data = data, kernel = "linear", cost = 10)


Parameters:
   SVM-Type:  C-classification
 SVM-Kernel:  linear
       cost:  10

Number of Support Vectors:  13

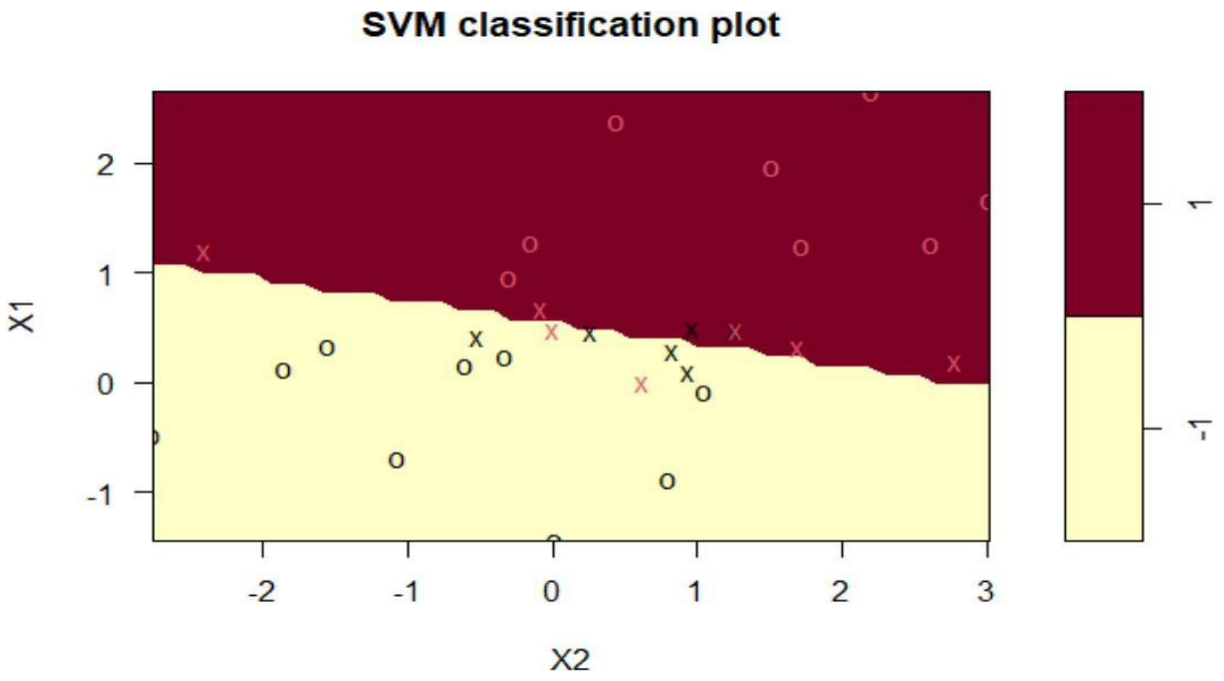
 ( 6 7 )


Number of Classes:  2

Levels:
 -1 1
```

Inference: From the above output there were 13 support vectors, 6 in one class and 7 in another class.

plot(svm,data)



Inference: The support vectors as crosses and the remaining observations are plotted as circles.

2. Example: Write R code to find the number of support vectors for classifying the data. Data as follows: 100 observations, in two columns and also get the SVM classification plot.

R Code with Output:

```
x = matrix(rnorm(100), 50, 2)
y = rep(c(-1, 1), c(25, 25))
x[y == 1,] = x[y == 1,] + 1
data = data.frame(x, y = as.factor(y))
svm = svm(y ~ ., data = data, kernel = "linear",cost=10)
summary(svm)
```

```
Call:
svm(formula = y ~ ., data = data, kernel = "linear",
    cost = 10)


Parameters:
   SVM-Type:  C-classification
 SVM-Kernel:  linear
       cost:  10

Number of Support Vectors:  34

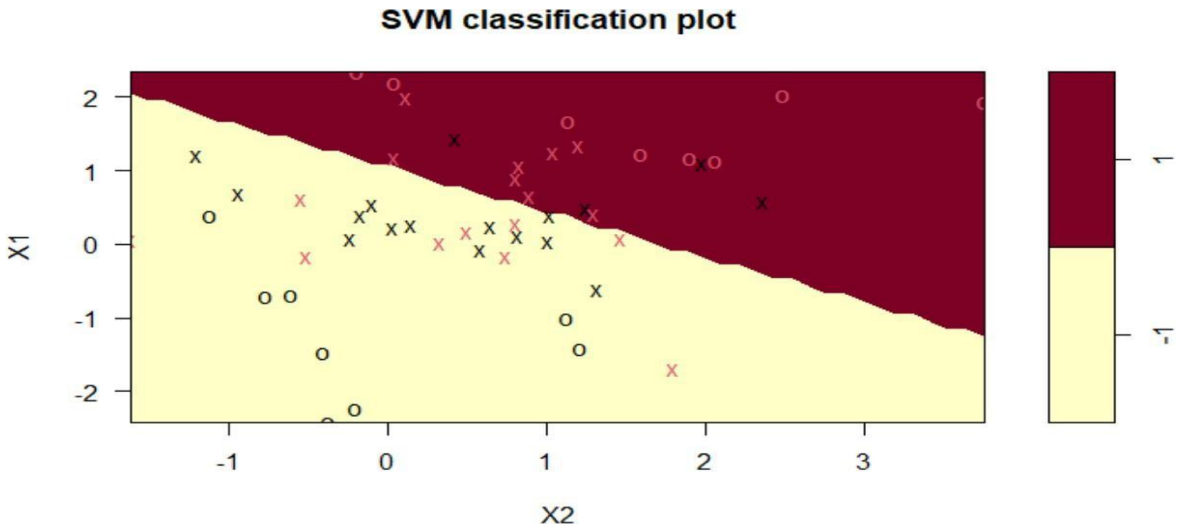
 ( 17 17 )


Number of Classes:  2

Levels:
 -1 1
```

Inference: From the above output there were 34 support vectors, 17 in one class and 17 in another class.

plot(svm,data)



Inference: The support vectors as crosses and the remaining observations are plotted as circles.

Viva Questions:

1 Define SVM?

Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems.

2. Define Hyperplane?

There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but we need to find out the best decision boundary that helps to classify the data points. This best boundary is known as the hyperplane of SVM.

3. Define support vectors?

The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector. Since these vectors support the hyperplane, hence called a Support vector.

9. Dimensionality reduction with PCA in R and K-means Clustering in R

9.1 Principal Component Analysis (PCA) in R

R Code:

Store the **Data** in Excel and Export the **Data**

PCA=prcomp(Data, scale=T)

summary(PCA)

1. Example: Obtain the percentage of principal components to the following data and interpret the results

A	38	35	36	38	35	30	38	30	26	32	35	36	31	38	35	30	29
B	61	61	62	65	60	61	61	62	60	58	56	57	58	65	60	62	52
C	33	33	65	67	59	33	33	56	60	62	61	60	52	67	59	56	59
D	32	32	27	34	43	32	32	44	48	52	33	39	42	34	43	44	56
E	50	50	31	33	36	50	50	35	31	30	32	29	30	33	36	35	31
F	38	35	54	57	70	30	38	72	77	49	50	51	56	57	70	72	52
G	65	60	65	74	60	62	65	62	61	68	58	62	61	74	60	62	63

R Code with Output:

Store the **Data** in Excel and Export the **Data**

PCA=prcomp(PCA_DATA,scale=T)

summary(PCA)

```
summary(PCA)
Importance of components:
                          PC1     PC2     PC3      PC4      PC5      PC6      PC7
Standard deviation     1.7831 1.4249 0.9095 0.79309 0.52442 0.18045 0.16253
Proportion of Variance0.4542 0.2901 0.1182 0.08986 0.03929 0.00465 0.00377
Cumulative Proportion 0.4542 0.7443 0.8624 0.95229 0.99157 0.99623 1.00000
```

Inference: PC1 explains 45% ,PC2 explains 29% ,PC3 explains 11% of the variation ,implies maximum information in the dataset out of seven variables can be encapsulated by these three principal components PC1,PC2,PC3.

2. Example: Obtain the percentage of principal components to the following data and interpret the results

X1	44	33	33	32	34	31	33	30	34	34	33	36	33	34	37
X2	1.3	2.2	2.2	2.6	2	1.8	2.2	3.6	1.6	1.5	2.2	1.7	2.2	1.3	2.6
X3	250	115	75	85	100	75	85	75	225	250	255	175	75	85	90
X4	0.63	0.59	0.56	0.55	0.54	0.59	0.56	0.46	0.63	0.6	0.63	0.58	0.55	0.57	0.62

R Code with Output:

PCA=prcomp(PCA_DATA, scale=T)

summary(PCA)

```
> PCA=prcomp(PCA_DATA, scale=T)
> summary(PCA)
Importance of components:
                          PC1    PC2    PC3     PC4
Standard deviation     1.6436 0.7171 0.7055 0.53547
Proportion of Variance 0.6753 0.1285 0.1244 0.07168
Cumulative Proportion  0.6753 0.8039 0.9283 1.00000
```

Inference: PC1 explains 67% of the total variance implies two-third of information in the dataset out of four variables can be encapsulated by one PC1. PC2 and PC3 explains 12 % each. From the above components, one can observe that first variable corresponding to the X_1 factor can be included in the model.

9.2 K-means Clustering in R

Algorithm:

1. Install appropriate package
2. Store the **Data** in Excel and Export the **Data**
3. Choose k value by any one of the three methods
4. Apply kmeans clustering to the given data.

R Code:

install.packages("factoextra") # To visualize the clusters
library(factoextra)

To choose k value: 3 methods are (Elbow, Silhouette, Gap)
fviz_nbclust(Data, kmeans, method = "wss", k.max = 8) #k.max argument is optional
or
fviz_nbclust(Data, kmeans, method = "silhouette", k.max = 8)
or
fviz_nbclust(Data, kmeans, method = "gap", k.max = 8)

#From the above 3 methods k value will be taken
clust=kmeans(Data, k value)

clust

fviz_cluster(clust, Data) #Displays cluster plot

1 Example: The following is the set of 10 data points. Cluster these points using R code in k means algorithm.

S.No.	1	2	3	4	5	6	7	8	9	10
X_1	18	22	25	32	38	42	44	46	52	60
X_2	12	18	25	28	32	38	40	43	45	50
X_3	18	22	35	38	39	43	50	55	60	65

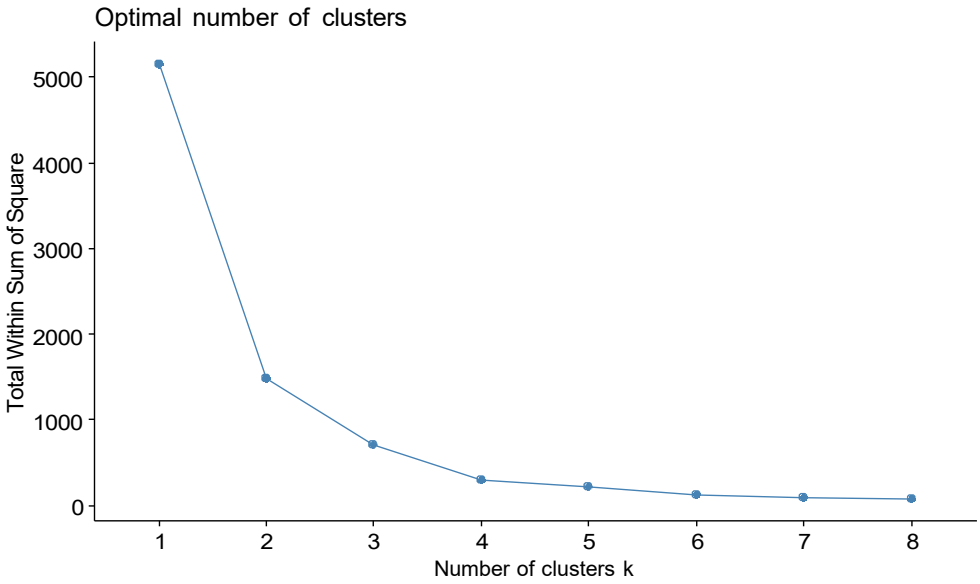
R Code with Output:

install.packages("factoextra")

library(factoextra)

fviz_nbclust(PCA_Data, kmeans, method = "wss", k.max = 8)

```
> library(factoextra)
> fviz_nbclust(PCA_Data, kmeans, method = "wss", k.max = 8)
```



Choose K=2

clust=kmeans(PCA_Data,2)

clust

```
> clust=kmeans(PCA_Data,2)
> clust
K-means clustering with 2 clusters of sizes 5, 5

Cluster means:
    X1   X2   X3
1 48.8 43.2 54.6
2 27.0 23.0 30.4

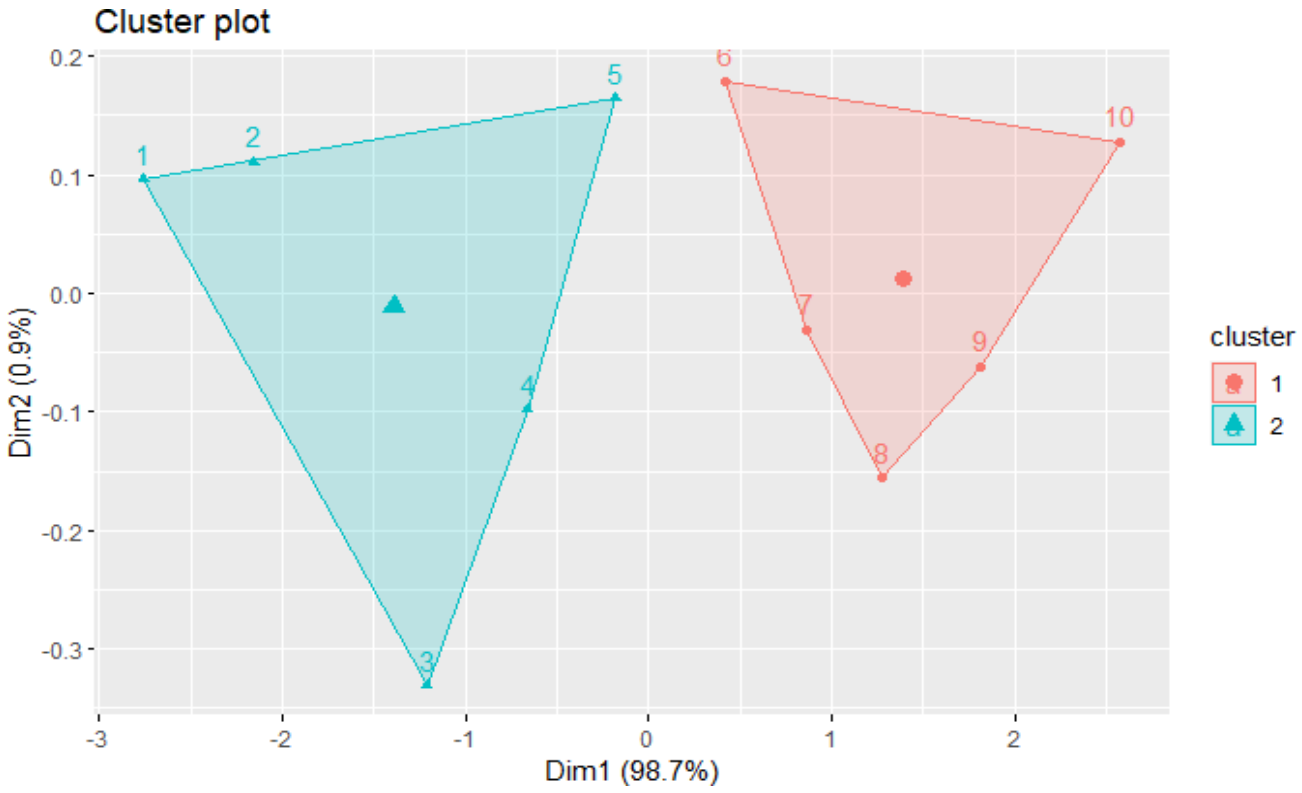
Clustering vector:
 [1] 2 2 2 2 2 1 1 1 1 1

Within cluster sum of squares by cluster:
[1] 592.8 889.2
 (between_SS / total_SS =  71.2 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss" "betweenss"
[7] "size"         "iter"         "ifault"
```

fviz_cluster(clust,PCA_Data)



2 Example: Cluster these data points using R code in k means algorithm.

X1	73	63	65	60	62	61	71	72	73	63	59	60	71	72	65
X2	60	62	61	63	60	56	65	64	60	62	62	62	65	64	55
X3	48	45	45	45	47	41	41	42	48	45	41	43	41	42	42
X4	38	44	74	70	68	42	32	35	38	44	35	30	32	35	56
X5	70	77	52	67	65	66	65	60	70	77	62	61	65	60	52
X6	65	63	62	67	65	64	58	62	65	63	60	56	58	62	62

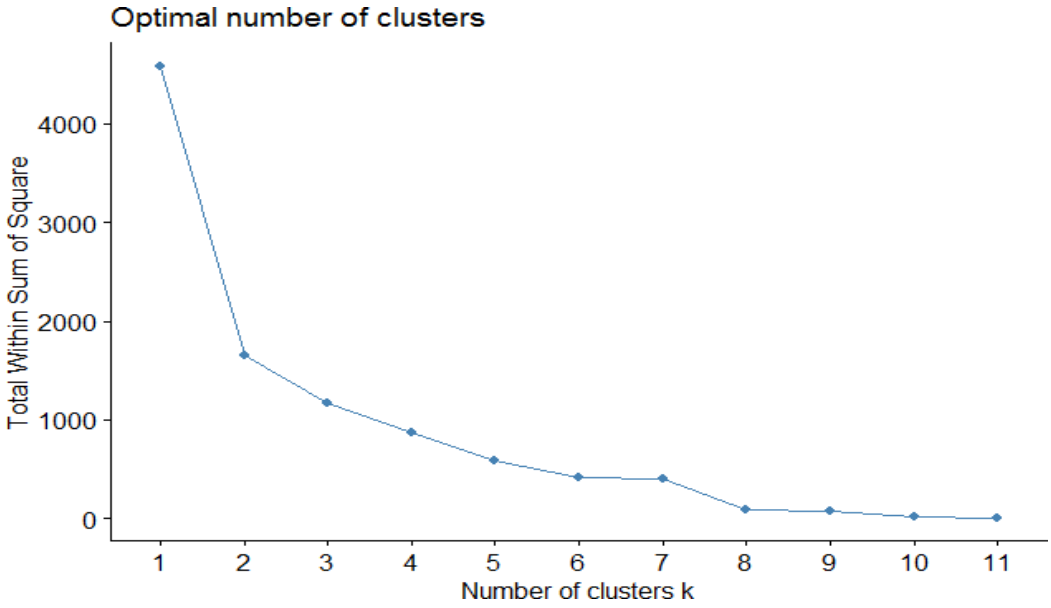
R Code with Output:

install.packages("factoextra")

library(factoextra)

fviz_nbclust(PCA_Data, kmeans, method = "wss", k.max = 11)

```
fviz_nbclust(DATA, kmeans, method = "wss", k.max = 11)
```



Choose K=2

clust=kmeans(PCA_Data,2)

clust

```
K-means clustering with 2 clusters of sizes 4, 11

Cluster means:
         Y1    Y2       Y3       Y4       Y5       Y6
1 63.00000 59.75 44.75000  67.00000 59.00000 64.00000
2 67.09091 62.00 43.36364  36.81818 66.63636 61.45455
```

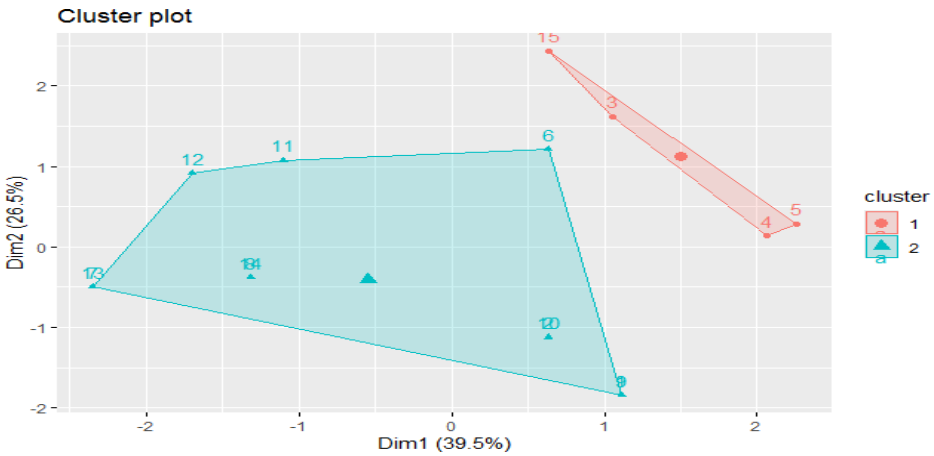
```
Clustering vector:
 [1] 2 2 1 1 1 2 2 2 2 2 2 2 2 2 1

Within cluster sum of squares by cluster:
[1]  461.500 1192.364
 (between_SS / total_SS =  63.9 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss
"
[6] "betweenss"    "size"         "iter"         "ifault"
```

```
fviz_cluster(clust,DATA)
```



Viva Questions:

1. Define Cluster.

   Cluster is defined as collections or groups of items with similar or different characteristics.

2. Define Principal Components.

   Principal component analysis (PCA) is a statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables (entities each of which takes on various numerical values) into a set of values of linearly uncorrelated variables called principal components.

3. What kind of Machine learning did PCA and Clustering techniques?

   PCA and Clustering techniques belongs to Un supervised Machine Learning .

10. Kernel SVM in R

Kernel SVM:

R Code:

```
install.packages("e1071")
library(e1071)
x = matrix(rnorm(32), 16, 2)          #Generating the data
y = rep(c(-1, 1), c(8, 8))            #Assigning +1 and -1 to the above data
x[y == 1,] = x[y == 1,] + 1           #bias the x variable when y =1
dat = data.frame(x, y = as.factor(y)) #representing the above data in table form using dataframe
svm = svm(y ~ ., data = dat, kernel = "radial",gamma=1,cost=1)
summary(svm)
plot(svm,dat)
```

1 Example: Write R code to find the number of support vectors for classifying the data in radial basis kernel. Data as follows: 400 observations, in two columns and also get the SVM classification plot.

R Code with Output:

R Code with Output:

```
x = matrix(rnorm(400), 200, 2)
y = rep(c(-1, 1), c(100, 100))
x[y == 1,] = x[y == 1,] + 1
data = data.frame(x, y = as.factor(y))
svm = svm(y ~ ., data = data, kernel = "radial",gamma=1, cost=1)
summary(svm)
```

```
Call:
svm(formula = y ~ ., data = data, kernel = "radial", gamma = 1,
    cost = 1)


Parameters:
   SVM-Type:  C-classification
 SVM-Kernel:  radial
       cost:  1

Number of Support Vectors:  134

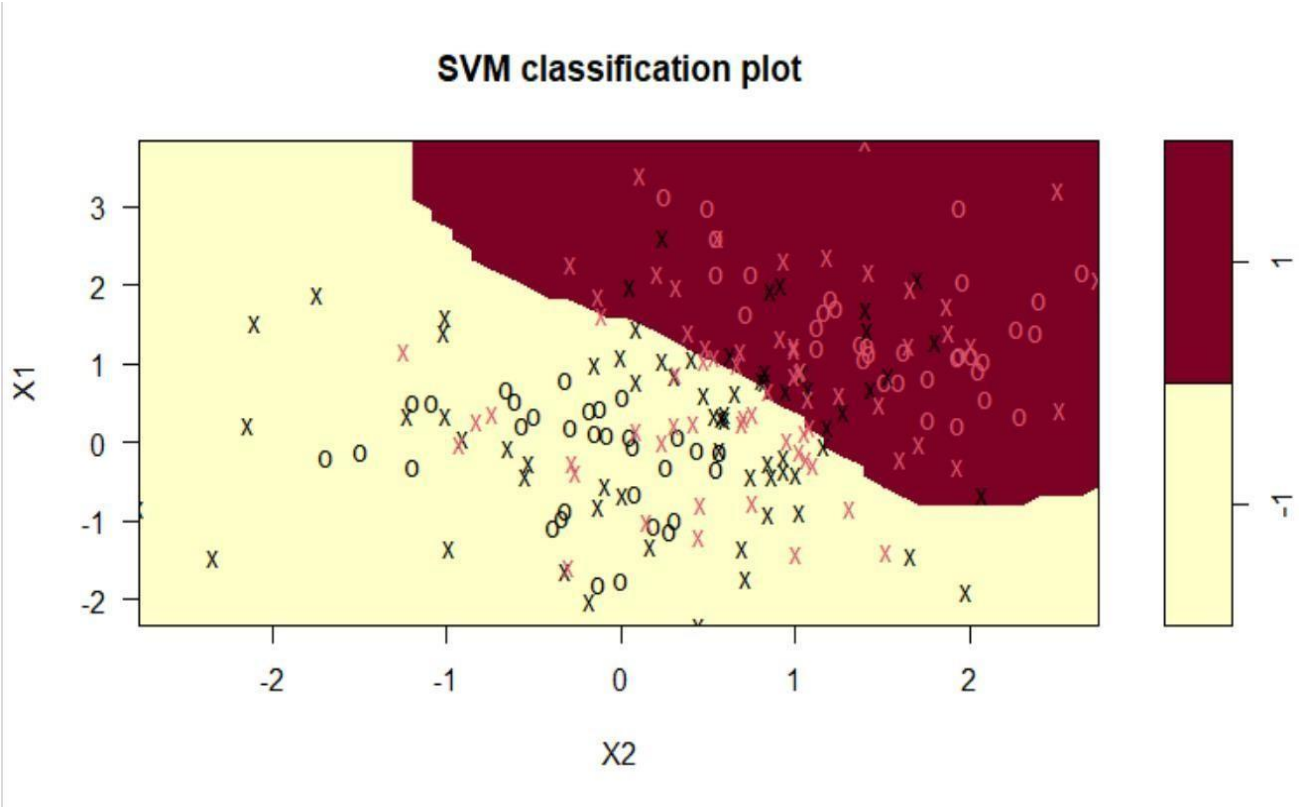
 ( 68 66 )


Number of Classes:  2

Levels:
 -1 1
```

Inference: From the above output there were 134 support vectors, 68 in one class and 66 in another class.

plot(svm,data)



2 Example: Write R code to find the number of support vectors for classifying the data in radial basis kernel. Data as follows: 300 observations, in two columns and also get the SVM classification plot.

R Code with Output:

R Code with Output:

```
x = matrix(rnorm(300), 150, 2)
y = rep(c(-1, 1), c(75, 75))
x[y == 1,] = x[y == 1,] + 1
data = data.frame(x, y = as.factor(y))
svm = svm(y ~ ., data = data, kernel = "radial",gamma=1, cost=10)
summary(svm)
```

```
Call:
svm(formula = y ~ ., data = data, kernel = "radial", gamma = 1, cost = 10)


Parameters:
   SVM-Type:  C-classification
 SVM-Kernel:  radial
       cost:  10

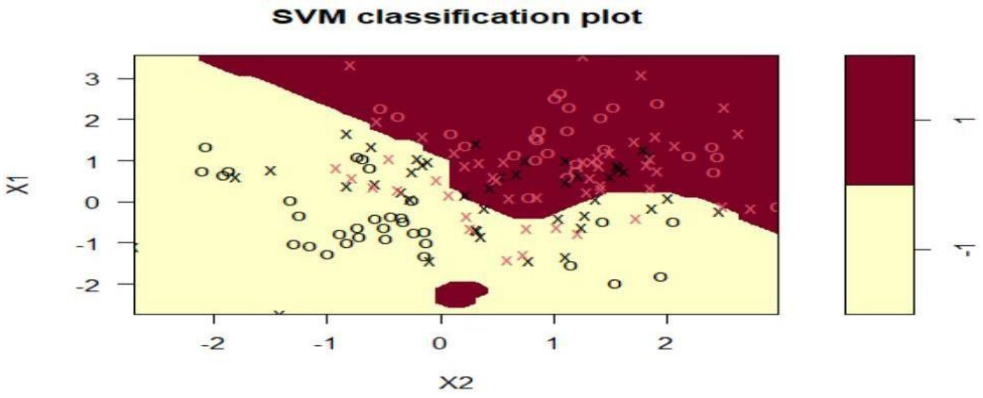
Number of Support Vectors:  91

 ( 43 48 )


Number of Classes:  2

Levels:
 -1 1
```

Inference: From the above output there were 91 support vectors, 43 in one class and 48 in another class.



Viva Questions:

1. Define Kernel.

Kernel Function is a method used to take data as input and transform it into the required form of processing data. "Kernel" is used due to a set of mathematical functions used in Support Vector Machine providing the window to manipulate the data.

2. What is the use of Kernel

In machine learning, a "kernel" is usually used to refer to the kernel trick, a method of using a linear classifier to solve a non-linear problem.

List of Additional programs

1. Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R programming

(i) Uniform Distribution

R Code:

runif(n, min, max)

1. Example: Generate 5 Uniform random numbers from 10 to 100.

R Code with Output:

runif(5,10,100)

```
[1] 42.61054 57.56261 23.57929 76.08863 25.25279
```

2. Example: Generate 10 random samples from the Uniform Distribution U(0,1)

R Code with Output:

runif(10,0,1)

```
[1] 0.67139102 0.34191665 0.50767938 0.89660762 0.05948337 0.65363865
[7] 0.83952941 0.12345681 0.97254724 0.75465741
```

(ii) Normal Distribution

R Code:

rnorm(n, mean, sd)

1. Example: Generate four random numbers from N (70, 10)

R Code with Output:

rnorm(4,70,sqrt(10))

```
[1] 74.84571 70.68275 67.49411 67.07436
```

2. Example: Generate 5 Random samples from the Normal distribution N (25,4)

R Code with Output:

rnorm(5,25,sqrt(4))

```
[1] 24.53206 24.90118 29.06580 23.51796 23.38991
```

(iii) Binomial Distribution

R Code:

rbinom(no of random numbers to be generate, n, p)

1. Example: Generate 6 Random samples from the Binomial distribution B (10, 0.5)

R Code with Output:

rbinom(6,10,0.5)

```
[1] 4 5 6 6 5 5
```

(iv) Exponential Distribution

R Code:

rexp(n, theta)

1. Example: Generate 5 Random samples from the Exponential distribution with parameter 10

R Code with Output:

rexp(5,10)

```
[1] 0.007909765 0.021077747 0.182569774 0.156448570 0.125258499
```

2. Fit linear regression to the given data

Hours: 1, 2, 4, 5, 5, 6, 6, 7, 8, 10, 11, 11, 12, 12, 14

Score: 64, 66, 76, 73, 74, 81, 83, 82, 80, 88, 84, 82, 91, 93, 89

R Code with output:

```
  df<-data.frame(hours=c(1, 2, 4, 5, 5, 6, 6, 7, 8, 10, 11, 11, 12, 12, 14),
       score=c(64, 66, 76, 73, 74, 81, 83, 82, 80, 88, 84, 82, 91, 93, 89))
head(df) # Displays the first 6 values in the data
   hours score
1     1    64
2     2    66
3     4    76
4     5    73
5     5    74
6     6    81

attach(df)
model <- lm(score~hours)
summary(model)

Call:
```

```
lm(formula = score ~ hours)

Residuals:
   Min     1Q Median     3Q    Max
-5.140 -3.219 -1.193  2.816  5.772

Coefficients:
             Estimate  Std. Error  t value      Pr(>|t|)
(Intercept)  65.334      2.106       31.023    1.41e-13  ***
hours        1.982       0.248       7.995     2.25e-06  ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.641 on 13 degrees of freedom
Multiple R-squared:  0.831,  Adjusted R-squared:  0.818
F-statistic: 63.91 on 1 and 13 DF,  p-value: 2.253e-06
```

INFERENCE:

In this model output intercept and hours variable are significant because its p-values are less than 0.05 so these are included in th modl equation.
The model equation is
Score =65.334+1.982*hours
The Multiple R-square value is 0.831 ,so the variable hours explains 83% variation in the dependent variable 'score'.
F –test is used to test the overall significant of the model
H_0: The model is not significant to the data
H_1:the model is significant to the data
The p-value for this f statistic is less than 0.05 so we reject H_0.Therefore the model is significant to the data.

3. Multiple linear regression

Testing overall hypothesis and testing significance of individual variables, model selections and prediction in R programming is called Multiple linear regression.

R Code:

lm(y ~ x1+x2+x3...,data)

Example 1: Fit a multiple linear regression to the following data set.

S.No.	1	2	3	4	5	6	7	8	9	10	11	12
X1	70	77	72	77	74	80	82	74	72	77	90	82
X2	100	105	112	105	125	112	129	125	112	105	140	129
X3	125	110	118	110	122	123	121	122	118	110	128	121
X4	65	70	66	70	58	68	61	58	66	70	82	61
Y	77	73	68	76	73	77	80	73	72	74	66	78

R Code with Output:

Left for student exercise.

Programme Outcomes

S No.	CO	Name of the Programme	Outcome of the Programme	Mention any Real-time application where we can apply this method	Which one is more efficient
1	CO1	Introduction to R Programming Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function) Acceptance/Rejection Sampling in R Maximum likelihood and method of moments estimation.	Perform built-in commands and operations with matrices in R-programming. Analyze the appropriate data type of variables and evaluate various Frequency Distributions in R - programming. Determine the mean and standard deviation of a sampling distribution using R.	R is a open source Software and it can be easily accessed. It represents the whole set of data by a single value. and gives us the location of central points.	
2	CO2	Testing of Hypothesis based on Z-test. Concept of p-value in R. Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R	Develop a code for testing of hypothesis using Z-test, t-test, chi-square-test, F-test. Classify Linear and Logistic regression.	To test the significance of hypothesis based on the p-value	
3	CO3	Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R Logistic Regression in R		To classify the objects into groups	
4	CO4	Classification with SVM in R Dimensionality reduction with PCA in R and K-means Clustering in R Kernel SVM in R	Write Code for SVM and Kernel SVM methods.	To classify the objects into groups when the data is large.	



...striving towards perfection